

Chest

CONTRIBUTIONS BY **Cheryl O. DuBose**, Ed.D., RT(R)(MR)(CT)(QM), MRSO

CONTRIBUTORS TO PAST EDITIONS Nancy Johnson, MEd, RT(R)(CV)(CT)(QM)(ARRT), FASRT, Karen Brown, RT(R), Kathy M. Martensen, BS, RT(R)

CONTENTS

Radiographic Anatomy

Chest, 70

Topographic Positioning Landmarks, 70

Parts of the Respiratory System, 71

Radiographic Positioning

Body Habitus, 78

Breathing Movements, 78

Positioning Considerations, 79

Radiation Protection, 79

Breathing Instructions, 80

Evaluation Criteria, 81

Digital Imaging Considerations, 86

Alternative Modalities and Procedures, 86

Clinical Indications, 86

Routine and Special Projections

PA Projection—Chest, 90

PA Projection—Chest (On Stretcher if Patient Cannot Stand), 91

Lateral Position—Chest, 92

Alternative Lateral Positions—Chest, 93

AP Projection—Chest, 94

Lateral Decubitus Position (AP Projection)—Chest, 95

AP Lordotic Projection—Chest, 96

Anterior Oblique Positions: RAO and LAO—Chest, 97

Posterior Oblique Positions: RPO and LPO—Chest, 99

Lateral Position—Upper Airway, 100

AP Projection—Upper Airway, 101

Radiographs for Critique, 102

RADIOGRAPHIC ANATOMY

Chest

Chest radiographic examinations are the most common of all radiographic procedures. Student radiographers typically begin their clinical experience taking chest radiographs. However, before beginning such clinical experience, it is important to learn and understand chest anatomy, including relative relationships of all anatomy within the chest cavity.

The **chest**, or **thorax**, is the upper portion of the trunk between the neck and the abdomen. Radiographic anatomy of the chest is divided into three sections: **bony thorax**, **respiratory system proper**, and **mediastinum**.

BONY THORAX

The **bony thorax** is the part of the skeletal system that provides a protective framework for the parts of the chest involved with breathing and blood circulation. **Thoracic viscera** is the term used to describe these parts of the chest, consisting of the lungs and the remaining thoracic organs contained in the mediastinum.

Anteriorly, the bony thorax consists of the **sternum** (breast-bone), which has three divisions. The superior portion is the **manubrium** (*mah-nu-bre-um*), the large center portion is the **body**, and the smaller inferior portion is the **xiphoid process**.

Superiorly, the bony thorax consists of the **2 clavicles** (collarbones) that connect the sternum to the **2 scapulae** (shoulder blades), the **12 pairs of ribs** that circle the thorax, and the **12 thoracic vertebrae** posteriorly (Fig. 2.1). A detailed description of all parts of the bony thorax is presented in Chapter 10.

Topographic Positioning Landmarks

Accurate and consistent radiographic positioning requires certain landmarks, or reference points, that can be used to center the image receptor (IR) correctly to ensure that all essential anatomy is included on that specific projection. These topographic landmarks should be parts of the body that are easily and consistently located on patients, such as parts of the bony thorax. For chest positioning, two of these landmarks are the **vertebra prominens** and the **jugular notch**.

VERTEBRA PROMINENS (SEVENTH CERVICAL VERTEBRA)

The vertebra prominens is an important landmark for determining the central ray (CR) location on a posteroanterior (PA) chest projection. It is the spinous process of the seventh cervical vertebrae (see Chapter 8). It can be palpated readily on most patients by applying light pressure with the fingertips at the base of the neck. The vertebra prominens is the first prominent process felt as you gently but firmly palpate down the back of the neck. With a little practice, this landmark can be located readily on most patients, especially with the neck in a flexed position.

JUGULAR NOTCH (MANUBRIAL OR SUPRASTERNAL NOTCH)

The jugular notch is an important landmark for determining the CR placement on anteroposterior (AP) chest projections. This is palpated easily as a deep notch or depression on the superior portion of the sternum below the thyroid cartilage.

The midthorax, at the level of T7 (seventh thoracic vertebra), can be located easily from these two landmarks, as described later in this chapter.

XIPHOID PROCESS (TIP)

The inferior tip of the sternum, the **xiphoid process**, which corresponds to the level of T9–T10, can also be palpated. The xiphoid process corresponds to the approximate level of the anterior portion of the diaphragm, which separates the chest cavity from the abdominal cavity. However, this is not a reliable landmark for positioning the chest because of variations in body habitus and the variable lower position of the posterior lungs, which may extend as far as T11 or T12 on inspiration, as shown in Fig. 2.2.

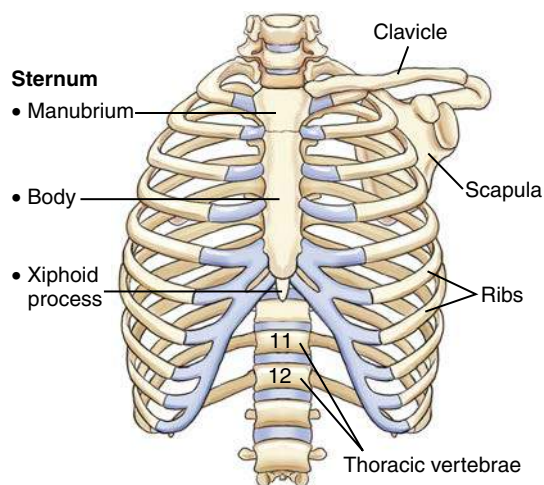


Fig. 2.1 Bony thorax.

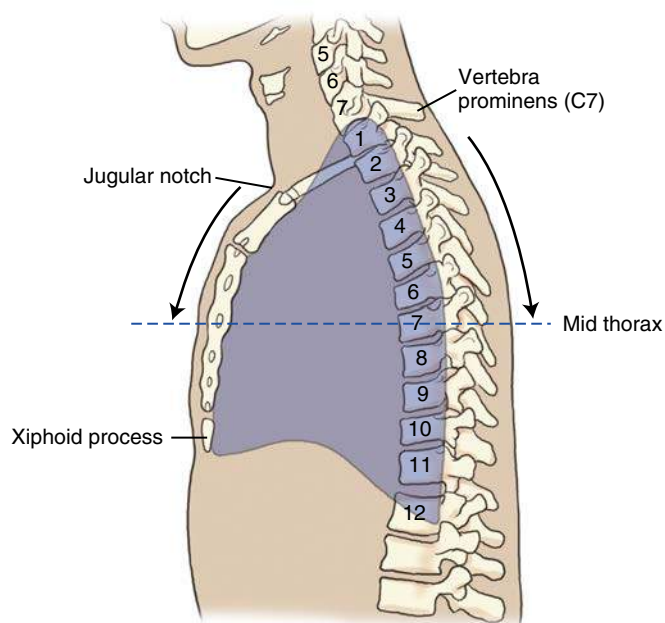


Fig. 2.2 Topographic landmarks.

PARTS OF THE RESPIRATORY SYSTEM

Respiration is the exchange of gaseous substances between the air we breathe and the bloodstream. The respiratory system consists of the parts of the body through which air passes as it travels from the nose and mouth into the lungs. Four general divisions of the respiratory system, shown in Fig. 2.3, are the **pharynx**, **trachea**, **bronchi**, and **lungs**.

An important structure of the respiratory system is the dome-shaped **diaphragm**, which is the primary muscle of inspiration. Each half of the diaphragm is called a **hemidiaphragm** ("hemi-" meaning half). As the dome of the diaphragm moves downward, it **increases** the volume of the thoracic cavity. This increase in volume, along with certain other dimensional movements of the thorax described later in this chapter, **decreases** the intrathoracic pressure, creating a "sucking" action or negative pressure effect, resulting in air being drawn into the lungs through the nose and mouth, pharynx, larynx, trachea, and bronchi. This causes the lungs to fill with air, which is known as inspiration.

PHARYNX

The **pharynx** (*far'-inks*) serves as a **passageway for food and fluids as well as air, making it common to the digestive and respiratory systems**. Air must pass through the pharynx before entering the rest of the respiratory system. The pharynx can be found in the posterior area between the nose and mouth above and the larynx and esophagus below.

The pharynx is approximately 5 inches (13 cm) in length and consists of three divisions, as shown in Fig. 2.4: **nasopharynx** (*na'-zo-far'-inks*), **oropharynx** (*o'-ro-far'-inks*), and **laryngopharynx** (*lah-ring'-go-far'-inks*). The superior portion of the pharynx is the nasopharynx, which is located posterior to the nose and houses the opening of the eustachian or auditory tube and the pharyngeal tonsils. The other end of the auditory tube is found in the middle ear and is useful in equalizing air pressure between the middle ear and the outside atmosphere.

The **hard palate** and the **soft palate** make up the roof of the oral cavity, separating the nasal cavity above from the mouth below. The lower posterior aspect of the soft palate is called the **uvula** (*u'-vu-lah*); this marks the boundary between the nasopharynx and the oropharynx. The oropharynx is found posterior to the mouth, with the tongue creating the anterior wall of the oropharynx. This section of the pharynx contains the palatine tonsils and the lingual tonsils.

The laryngopharynx lies above and posterior to the larynx (voice box) and extends from the upper border of the **epiglottis** (*ep'-i-glot'-is*) to where the laryngopharynx narrows to join the esophagus.

The upper portion of the epiglottis projects upward behind the tongue and acts as a lid for the slanted opening of the larynx. During the act of swallowing, the epiglottis flips down and covers the laryngeal opening, preventing food and fluid from entering the larynx and bronchi.

Additional structures shown on this sectional lateral drawing are the **hyoid bone**, **thyroid cartilage** of the larynx (Adam's apple), **thyroid gland**, and **trachea**, which are described in greater detail in the subsequent sections on the larynx and the trachea.

Esophagus

The **esophagus** is the part of the digestive system that connects the pharynx with the stomach. Note the relationship of the esophagus to both the pharynx and the larynx. It begins at the distal end

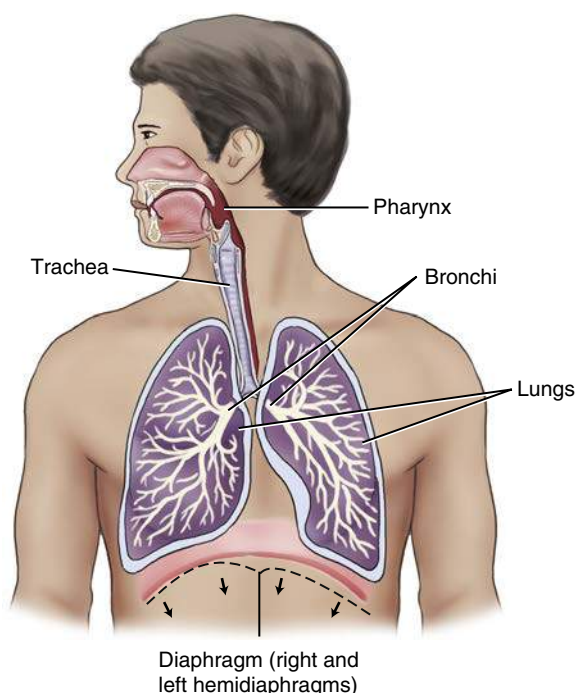


Fig. 2.3 Respiratory system.

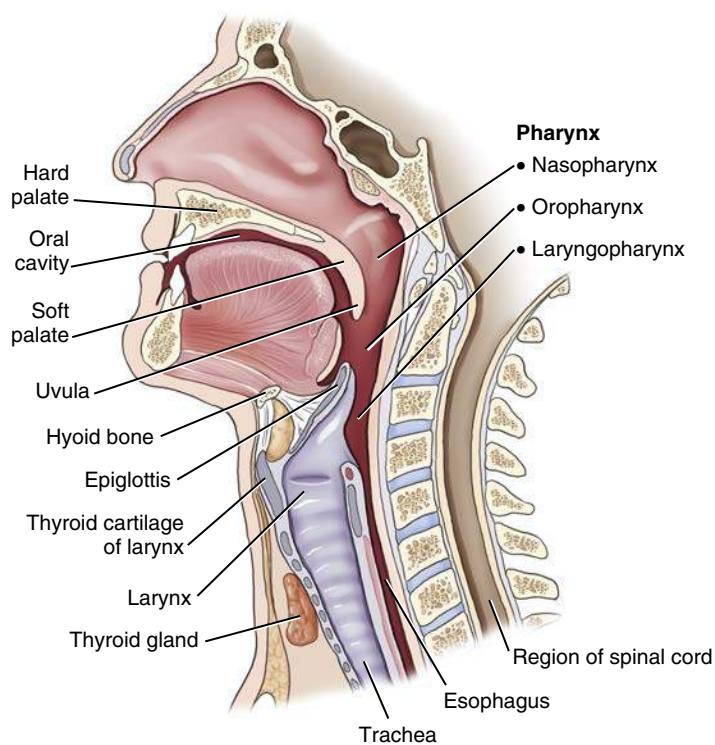


Fig. 2.4 Pharynx, upper airway (midsagittal section).

of the laryngopharynx and continues downward to the stomach, **posterior to the larynx and trachea.** (Chapter 12 describes the esophagus, along with the upper gastrointestinal [UGI] system, in detail.)

Larynx (Voice Box)

The **larynx** is a cage-like, cartilaginous structure that is approximately 1½ to 2 inches (4 to 5 cm) in length in an adult. The larynx is located in the anterior portion of the neck, suspended from a small bone called the **hyoid** (Fig. 2.5). The hyoid bone is found in the upper neck just below the tongue or floor of the mouth (see Fig. 2.4). The hyoid bone is *not* part of the larynx.

The larynx serves as the organ of voice. Sounds are made as air passes between the **vocal cords** located within the larynx (Fig. 2.6). The upper margin of the larynx is at the approximate level of **C3** (third cervical vertebra). Its lower margin, where the larynx joins with the trachea, is at the level of **C6**.

The framework of the larynx consists of cartilages that are connected by ligaments and moved by numerous muscles that assist in the complex sound-making or voice process. The largest and least mobile of these cartilages is the **thyroid cartilage**, which consists of two fused plate-like structures that form the anterior wall of the larynx. The prominent anterior projection of the thyroid cartilage is palpated easily and is known as the **laryngeal prominence**, or Adam's apple. This prominent structure is an important positioning landmark because it is easy to locate. Generally larger in males than females, the laryngeal prominence of the thyroid cartilage is located at the level of C4–C5 and is an excellent topographic reference for locating specific skeletal structures in this region.

The **cricoid** (*kri'-koid*) **cartilage** is a ring of cartilage that forms the inferior and posterior wall of the larynx. It is attached to the first ring of cartilage of the trachea.

One of the cartilages that make up the larynx is the uniquely shaped **epiglottis**, which resembles a leaf with the narrow distal stem portion attached to a part of the thyroid cartilage. As described previously, the epiglottis flips down and covers the trachea during the act of swallowing (see arrow in Fig. 2.6).

Sectional Image of Larynx Because of the wide acceptance of computed tomography (CT) and magnetic resonance imaging (MRI), the technologist must recognize anatomic structures in sectional images. Fig. 2.7 shows an axial CT view of the midportion of the larynx at the approximate level of C5. Only major structures are labeled in this section.

NOTE: CT images such as those seen here often are viewed as though one were standing at the patient's feet, looking toward the head. Thus, the patient's right is to the viewer's left. This is the same way that radiographs are reviewed. (see Chapter 1).

TRACHEA

Connecting the larynx to the main bronchi is the **trachea**, or wind-pipe. It is a fibrous muscular tube about ¾ inch (2 cm) in diameter and 4½ inches (11 cm) long. Approximately 16 to 20 C-shaped rings of cartilage are embedded in its anterior wall. These rigid rings keep the airway open by preventing the trachea from collapsing during expiration.

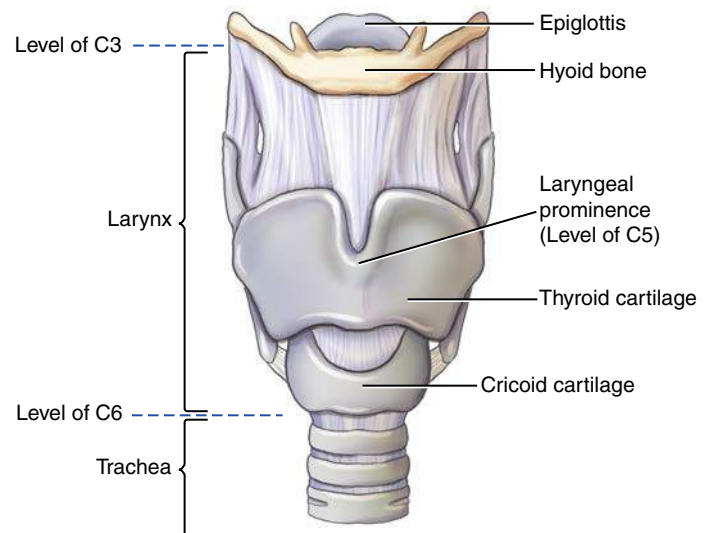


Fig. 2.5 Larynx (frontal view).

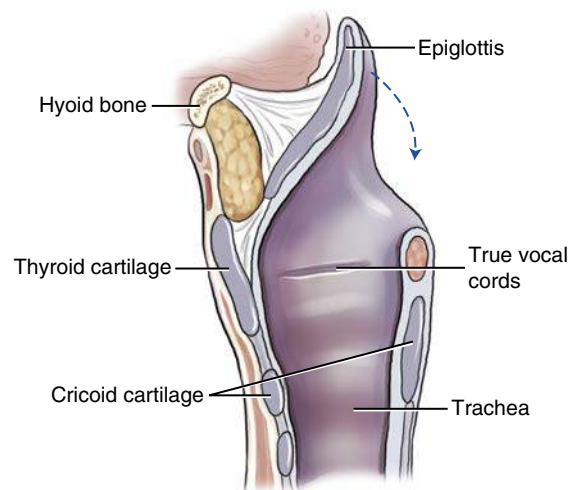


Fig. 2.6 Larynx (lateral view).

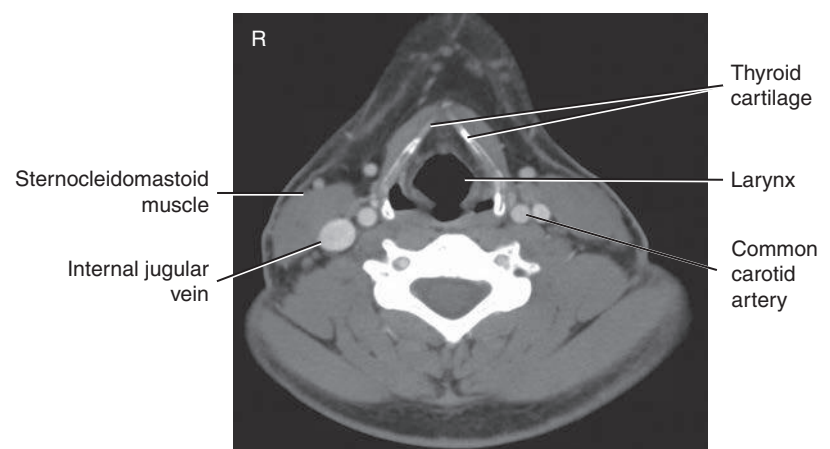


Fig. 2.7 CT image of neck through larynx—axial section at level of C5.

The trachea, located just anterior to the esophagus, extends from its junction with the larynx at the level of **C6** downward to the level of **T4** or **T5** (fourth or fifth thoracic vertebra), where it divides at the **carina** (last tracheal cartilage) into the right and left primary bronchi.

The endocrine system glands typically imaged with the respiratory system include the **thyroid**, **parathyroid**, and **thymus glands**.

Thyroid Gland

The thyroid gland is located anteriorly in the neck region just below the thyroid cartilage, with its right and left lobes lying on each side of the trachea (Fig. 2.8). In an adult, it weighs about 1 ounce (25 to 30 g) and is more radiosensitive than most body structures or organs. It is important for technologists to reduce exposure to this region as much as possible by shielding and collimation of the x-ray beam.

A unique feature of the thyroid gland is its ability to store certain hormones and release them slowly to aid in the regulation of body metabolism. These hormones also help to regulate body growth and development, especially in children. The thyroid gland serves as a counterweight to the parathyroid glands, stimulating the increased deposition of calcium in the bone, thereby lowering blood calcium levels.

Parathyroid Glands

Parathyroid glands are small, round glands that are embedded in the posterior surface of each lobe of the thyroid gland. Usually, two parathyroids are attached to each lateral thyroid lobe (four total), as shown in Fig. 2.8. These glands store and secrete hormones that aid in specific blood functions, including maintenance of blood calcium levels by stimulating bone breakdown to increase calcium in the blood.

Thymus Gland

The thymus gland is located inferior to the thyroid gland and anterior and superior to the heart (see Fig. 2.8). It is described later in this chapter as part of the mediastinal structures (see Fig. 2.22).

RADIOGRAPHS

AP and lateral radiographs of the upper airway allow visualization of the air-filled trachea and larynx. The AP radiograph in Fig. 2.9 shows a column of air primarily in the upper trachea region, as is seen in the lower half of the radiograph (darkened area, arrows). Certain enlargements or other abnormalities of the thymus or thyroid glands can be demonstrated on such radiographs, as can pathology within the airway system itself.

The lateral radiograph (Fig. 2.10) shows the air-filled trachea and larynx (A) and the region of the esophagus (B) and shows the locations relative to each other. The esophagus is located posteriorly in relation to the trachea. The general locations of the thyroid gland (C) and the thymus gland (D) are identified.

Sectional Image of the Trachea

Fig. 2.11 is a CT image through the upper chest at the approximate level of T3. Observe that the trachea is located anteriorly to the esophagus and that both of these are anterior to the thoracic vertebrae. The upper lungs are located to each side of the trachea and the thoracic vertebrae.

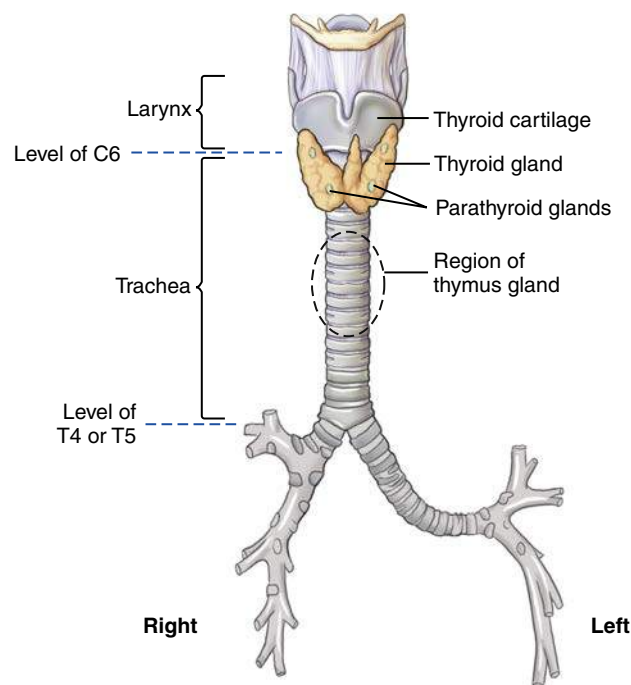


Fig. 2.8 Trachea.

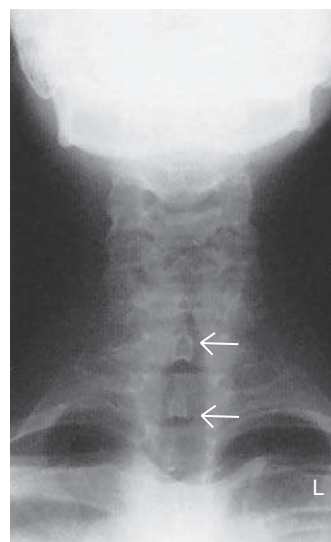


Fig. 2.9 AP upper airway.

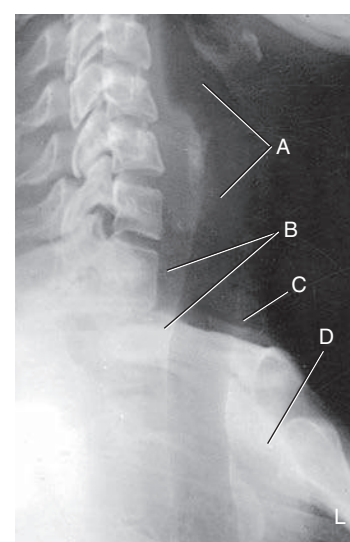


Fig. 2.10 Lateral upper airway. A, Air-filled trachea and larynx. B, Esophagus. C, Region of the thyroid gland. D, Region of the thymus gland.

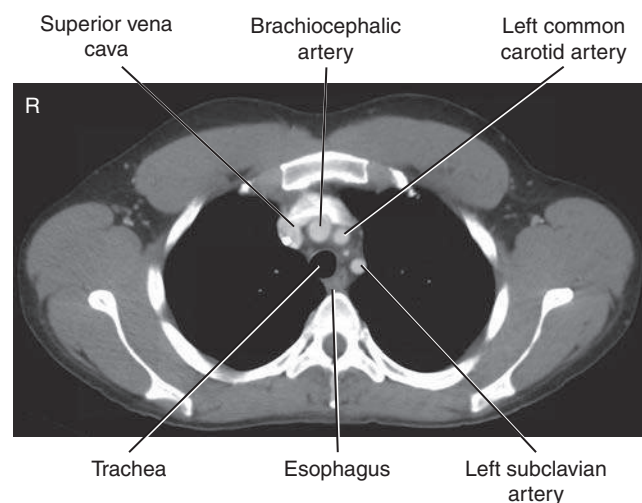


Fig. 2.11 Axial section at level of T3.

Branches of Aortic Arch

The major arterial branches from the aortic arch are identified in Fig. 2.11. The major branches include the brachiocephalic, left common carotid, and left subclavian arteries. The superior vena cava is a large vein draining blood from the head, neck, and upper limbs and returning it to the heart.

RIGHT AND LEFT BRONCHI

The trachea bifurcates at the carina to form the **right** and **left primary bronchi**, also known as the right and left main stem bronchi.

The **right primary bronchus** is **wider** and **shorter** than the left bronchus. The right primary bronchus is also more **vertical**; therefore, the angle of divergence from the distal trachea is less abrupt for the right bronchus than for the left. This **difference in size and shape** between the two primary bronchi is important because food particles or other foreign objects that happen to enter the respiratory system are more likely to enter and lodge in the **right** bronchus.

The **carina** (*kah-ri'-nah*) is a specific prominence, or ridge, of the lowest tracheal cartilage and marks the division of the trachea into the right and left bronchi (Fig. 2.12). As viewed from above through a bronchoscope, the carina is to the left of the midline, and the right bronchus appears more open than the left, demonstrating the likelihood that particles descending the trachea will enter the right bronchus.

Secondary Bronchi, Lobes, and Alveoli

In addition to differences in size and shape between the right and left bronchi, another important difference is that the **right** bronchus divides into **three** secondary bronchi, but the **left** divides into only **two**, with each entering individual lobes of the lungs (Fig. 2.13). The **right lung** contains **three lobes**, and the **left lung** contains **two lobes**, as is demonstrated in Figs. 2.14 and 2.15. These secondary bronchi continue to subdivide into smaller branches, called **bronchioles**, that spread to all parts of each lobe.

Each of these small **terminal bronchioles** terminates in very small air sacs called **alveoli**. The two lungs contain approximately 500 million to 700 million alveoli. Oxygen and carbon dioxide are exchanged in the blood through the thin walls of the alveoli.

Sectional Image of Bronchi and Lungs Fig. 2.14 presents an axial (sectional) drawing through the heart at the approximate level of T7.

LUNGS

The fourth and last division of the respiratory system comprises the two large, spongy **lungs**, which are located on each side of the thoracic cavity. The lungs fill all of the space not occupied by other structures. The right lung is made up of **three lobes**—the **superior** (upper), **middle**, and **inferior** (lower) lobes—divided by **two deep fissures**. The oblique fissure separates the inferior and middle lobes, whereas the horizontal fissure separates the superior and middle lobes. The left lung has only **two lobes**—the **superior** (upper) and **inferior** (lower)—separated by a **single deep oblique fissure**.

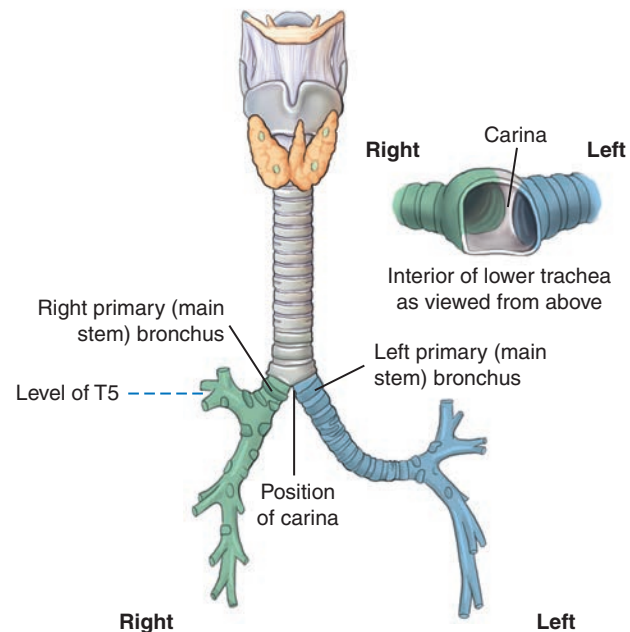


Fig. 2.12 Bronchi.

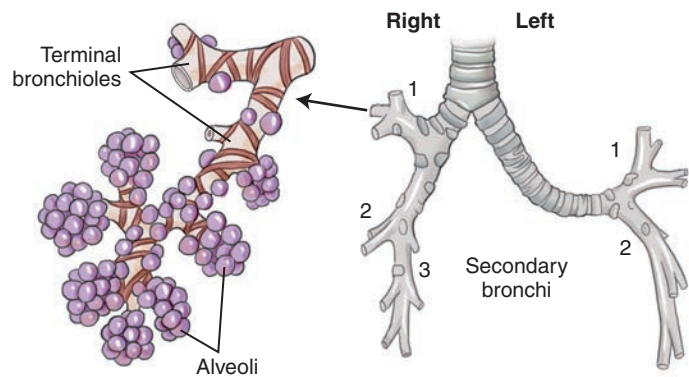


Fig. 2.13 Secondary bronchi and alveoli.

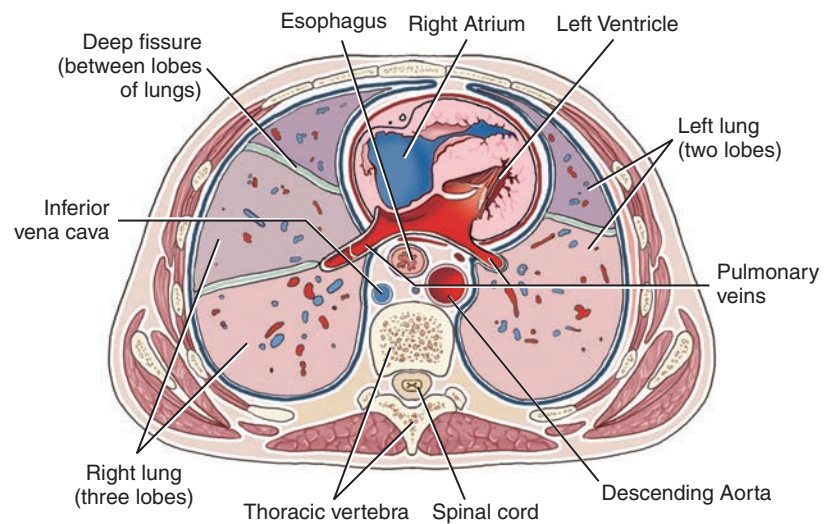


Fig. 2.14 Axial (sectional) image of lungs and heart, level of T7.

The lungs are composed of a light, spongy, highly elastic substance called **parenchyma** (*pah-reng'-ki-mah*). This substance allows for the breathing mechanism responsible for expansion and contraction of the lungs, which brings oxygen into and removes carbon dioxide from the blood through the thin walls of the alveoli.

Each lung is contained in a delicate double-walled sac, or membrane, called the **pleura**, which can be visualized in both frontal (see Fig. 2.15) and sectional (Fig. 2.16) drawings. The outer layer of this pleural sac lines the inner surface of the chest wall and diaphragm and is called the **parietal pleura**. The inner layer that covers the surface of the lungs, also dipping into the fissures between the lobes, is called the **pulmonary** or **visceral pleura** (Fig. 2.16).

The potential space between the double-walled pleura, called the **pleural cavity**, contains a lubricating fluid that allows movement of one or the other during breathing. When a lung collapses, or when air or fluid collects between these two layers, this space may be visualized radiographically. Air or gas present in this pleural cavity results in a condition called a **pneumothorax**, in which air or gas pressure in the pleural cavity may cause the lung to collapse. Accumulation of blood in the pleural cavity creates a condition called a **hemothorax**, whereas fluid within the cavity is referred to as **pleural effusion**.

Sectional Drawing of Lungs and Heart

Fig. 2.16 presents an axial sectional drawing through the lower third of the mediastinum and lungs. The double-walled membrane, the **pleura**, which completely encloses the lungs, including around the heart, is clearly shown. The outer membrane, the **parietal pleura**, and the inner membrane, the **pulmonary** (or **visceral**) **pleura**, are clearly visible, as is the potential space between them, the **pleural cavity**.

The double-walled **pericardial sac**, which surrounds the heart, is also identified. This drawing shows the relationship of the pericardial sac surrounding the heart with the pleural sac surrounding the lungs. The pleural and pericardial spaces or cavities are exaggerated on this drawing to show these parts better. Normally, no space exists between the double walls of the pericardial sac or between the parietal and visceral pleura unless pathology is present.

Sectional Image

The CT image in Fig. 2.17 at the approximate level of T10 demonstrates the relationship and relative size of the heart, descending aorta, esophagus, and lungs. The heart is located more to the **left**, as can be seen on a PA chest radiograph. The heart is also shown to be located in the anterior portion of the chest cavity directly behind the sternum and left anterior ribs. The esophagus is posterior to the heart, with the descending aorta between the esophagus and the thoracic vertebrae.

PA Chest Radiograph

An enormous amount of medical information can be obtained from a properly exposed and carefully positioned PA chest radiograph. Although the technical factors are designed for optimal visualization of the lungs and other soft tissues, the bony thorax can also be seen. The clavicles, scapulae, and ribs can be identified through careful study of the chest radiograph in Fig. 2.18. The sternum and thoracic vertebrae are superimposed along with mediastinal structures, such as the heart and great vessels;

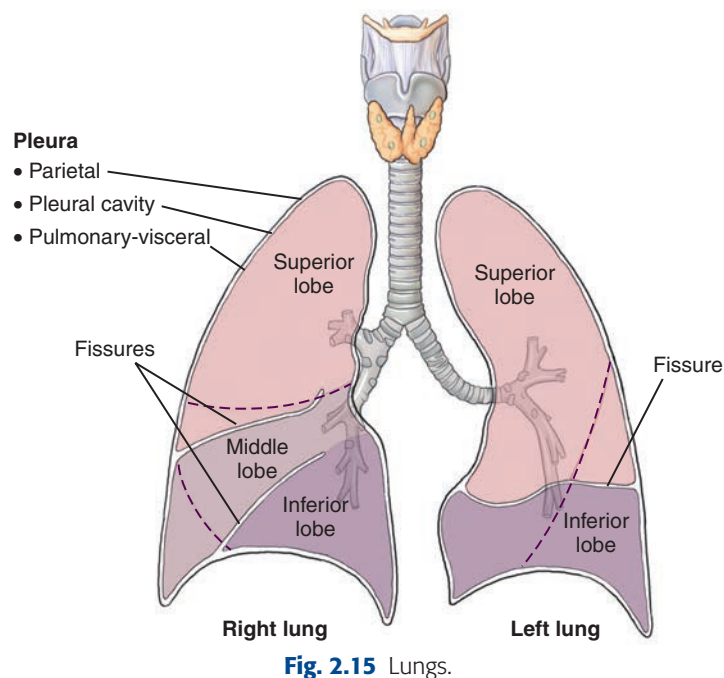


Fig. 2.15 Lungs.

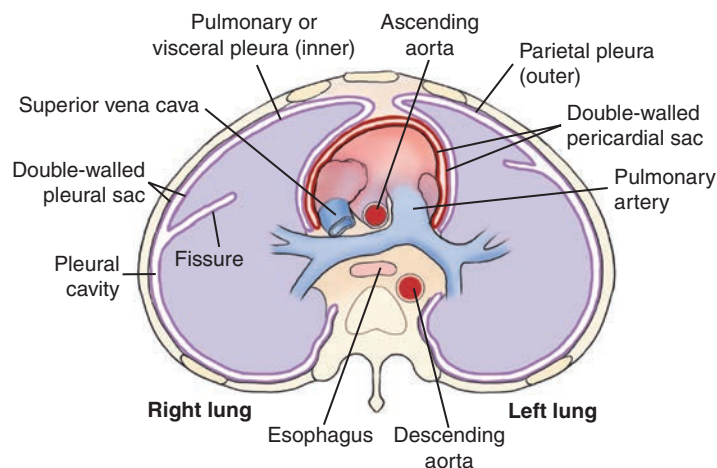


Fig. 2.16 Cross-section of lower mediastinum and lungs.

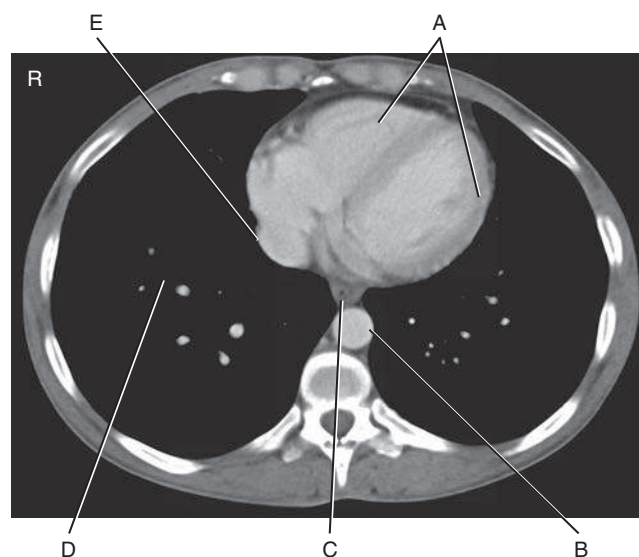


Fig. 2.17 CT image of lower thorax (level of T10). A, Heart. B, Descending aorta. C, Esophagus. D, Right lung. E, Inferior vena cava.

therefore, the sternum and vertebrae are not well visualized on a PA chest radiograph.

The **lungs** and the **trachea** (see Fig. 2.18, dotted outline, A) of the respiratory system are well shown, although usually the primary bronchi are not seen easily. The larynx is usually located above the top border of the radiograph and cannot be seen. The heart, large blood vessels, and diaphragm are well visualized.

The labeled parts of the radiograph are demonstrated in Fig. 2.19, a frontal view of the thorax with the bony structures removed. The thyroid gland, large blood vessels, and thymus gland are shown in relation to the lungs and heart.

Parts of lungs Radiographically important parts of the lungs, most also shown in the accompanying drawing, (see Figs. 2.18 and 2.19) include the following:

The **apex** (B) of each lung is the **rounded upper area above the level of the clavicles**. The apices of the lungs extend up into the lower neck area to the level of T1 (first thoracic vertebra). This important part of the lungs must be included on chest radiographs.

The **carina** (C) is shown as the point of bifurcation, the lowest margin of the separation of the trachea into the right and left bronchi.

The **base** (D) of each lung is the lower concave area of each lung that rests on the **diaphragm** (E). The diaphragm is a muscular partition that separates the thoracic and abdominal cavities.

The **costophrenic angle** (F) refers to the extreme outermost lower corner of each lung, where the diaphragm meets the ribs. When positioning for chest radiographs, you should know the relative locations of the uppermost and lowermost parts of the lungs—the apices and the costophrenic angles, respectively—to ensure that these regions are included on every chest radiograph. Pathology, such as a small amount of fluid collection, would be evident at these costophrenic angles in the erect position.

The **hilum** (hilus) (G), also known as the **root** region, is the central area of each lung, where the bronchi, blood vessels, lymph vessels, and nerves enter and leave the lungs.

Lateral chest view The lateral chest radiograph (Fig. 2.20) is marked to show the same parts as those labeled in the accompanying drawing (Fig. 2.21). This drawing depicts the left lung as seen from the medial aspect. Because this is the left lung, only two lobes are seen. Some of the **lower lobe** (D) extends above the level of the **hilum** (C) posteriorly, whereas some of the **upper lobe** (B) extends below the hilum anteriorly. The posterior portion of the diaphragm is the most inferior part of the **diaphragm**. The single deep **oblique fissure** that divides the two lobes of the left lung is shown again, as is the end-on view of a bronchus in the hilar region.

The right lung is usually about 1 inch (2.5 cm) shorter than the left lung. The reason for this difference is the large space-occupying liver, which is located in the right upper abdomen and which pushes up on the **right hemidiaphragm**. The right and left hemidiaphragms are seen on the lateral chest radiograph (Fig. 2.20F). The more superior of the two is the right hemidiaphragm, which is also seen on the PA chest radiograph (see Fig. 2.18).

Mediastinum

The medial portion of the thoracic cavity between the lungs is called the **mediastinum**. The thyroid and parathyroid glands, as described

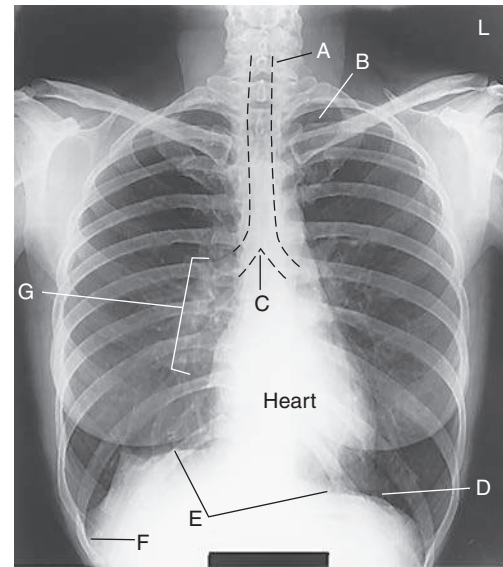


Fig. 2.18 PA chest radiograph.

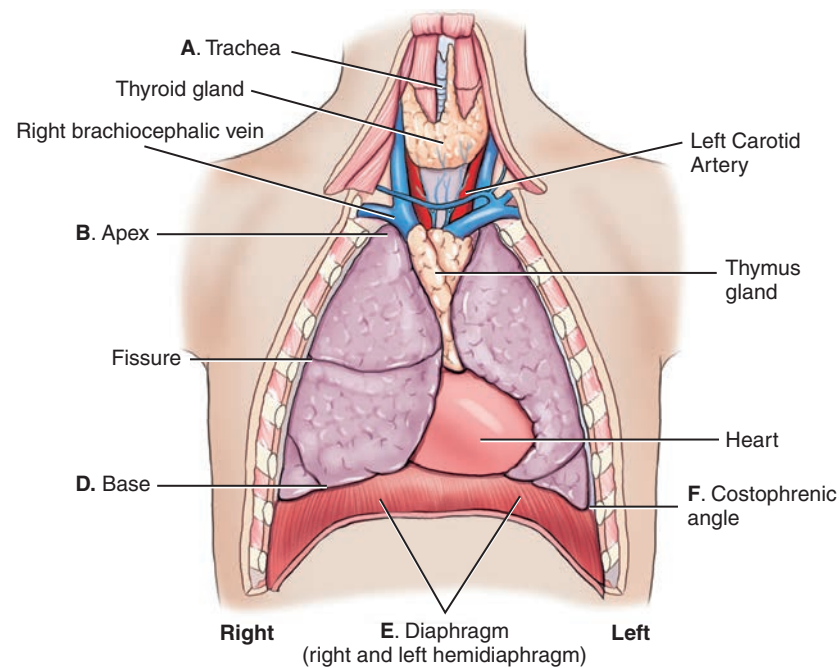


Fig. 2.19 Lungs and mediastinum.

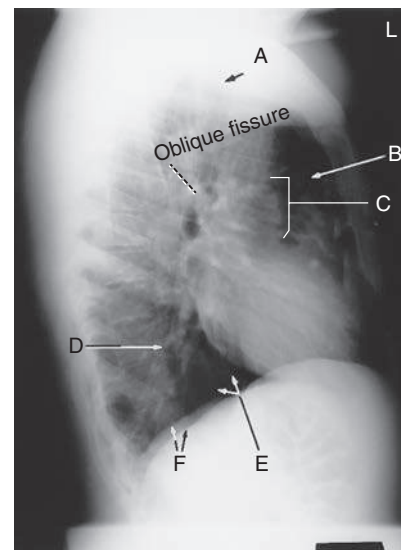


Fig. 2.20 Lateral chest radiograph.

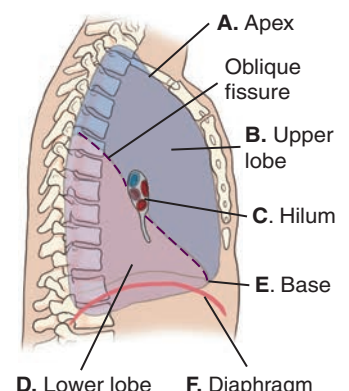


Fig. 2.21 Medial left lung.

earlier in this chapter, are *not* considered mediastinal structures because they are located more superiorly and are not within the mediastinum. However, the thymus gland is located within the mediastinum, inferior to the thyroid gland and anterior to the trachea and esophagus (Fig. 2.22).

Four radiographically important structures located in the mediastinum are the (1) **thymus gland**, (2) **heart and great vessels**, (3) **trachea**, and (4) **esophagus**.

Thymus gland The **thymus gland**, located behind the upper sternum, is said to be a temporary organ because it is very prominent in infancy and reaches its maximum size of about 40 g at puberty, then gradually decreases in size until it almost disappears in adulthood. At its maximum size, it would appear much larger than the organ shown in Fig. 2.22. It may be visualized on chest radiographs of children but generally is not seen in adult radiographs because the denser lymphatic tissue has been replaced by less dense fatty tissue. At its maximum development, the thymus gland lies above and anterior to the heart and pericardium.¹

The thymus gland has a large role in the development of the immune system that helps the body resist disease. It is essential to the growth and development of thymic lymphocytes or T cells, which serve in rejecting things foreign to the body.

Heart and great vessels The **heart** and the roots of the **great vessels** are enclosed in a double-walled sac called the pericardial sac (shown in Fig. 2.16). The heart is located posterior to the body of the sternum and anterior to T5 to T8. It lies obliquely in the mediastinal space, and approximately two-thirds of the heart lies to the left of the median plane.

The **great vessels** in the mediastinum are the inferior vena cava and superior vena cava, aorta, and large pulmonary arteries and veins. The **superior vena cava** is a large vein that returns blood to the heart from the upper half of the body (see Fig. 2.22). The **inferior vena cava** is a large vein that returns blood from the lower half of the body.

The **aorta** is the largest artery in the body (2 to 3 cm in diameter in an average adult). It carries blood to all parts of the body through its various branches. The aorta is divided into three parts: **ascending aorta** (coming up out of the heart); **arch of the aorta**; and **descending aorta**, which passes through the diaphragm into the abdomen, where it becomes the abdominal aorta.

Various **pulmonary arteries and veins** present in the mediastinum are shown in Figs. 2.23 and 2.24. These supply blood and return blood to and from all segments of the lungs. The capillary network surrounds the small air sacs, or alveoli, where oxygen and carbon dioxide are exchanged with the blood through the thin-walled air sacs.

See Chapter 17 for more complete drawings of the heart and great vessels as part of the total-body circulatory system.

Trachea and esophagus The trachea, within the mediastinum, separates into the right and left primary and secondary bronchi, as shown in Fig. 2.23.

The proximal esophagus is located posterior to the trachea and continues down through the mediastinum **anterior to the descending aorta** until it passes through the diaphragm into the stomach.

Note in Fig. 2.24 that the heart is located in the **anterior** aspect of the thoracic cavity, directly behind the sternum.

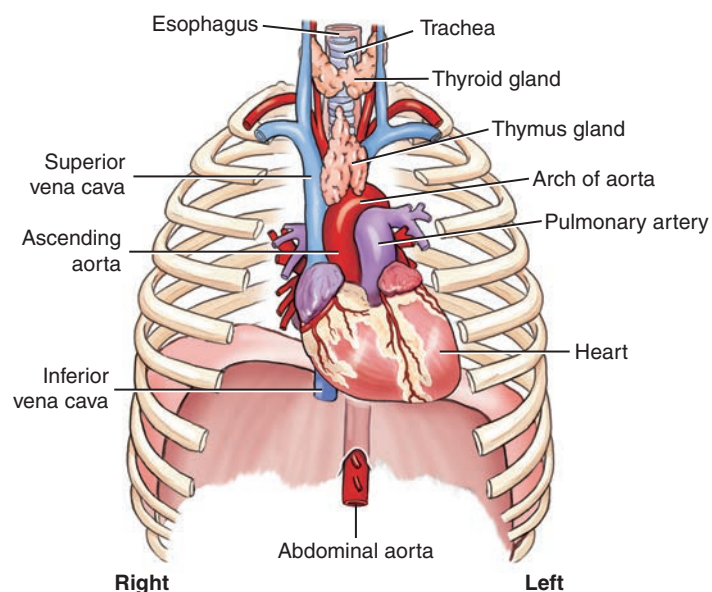


Fig. 2.22 Structures within mediastinum (anterior view).

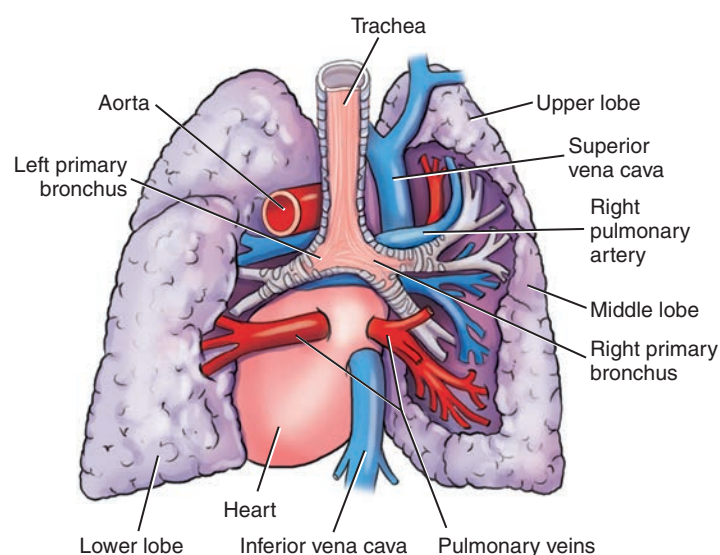


Fig. 2.23 Lungs and structures within mediastinum (posterior view).

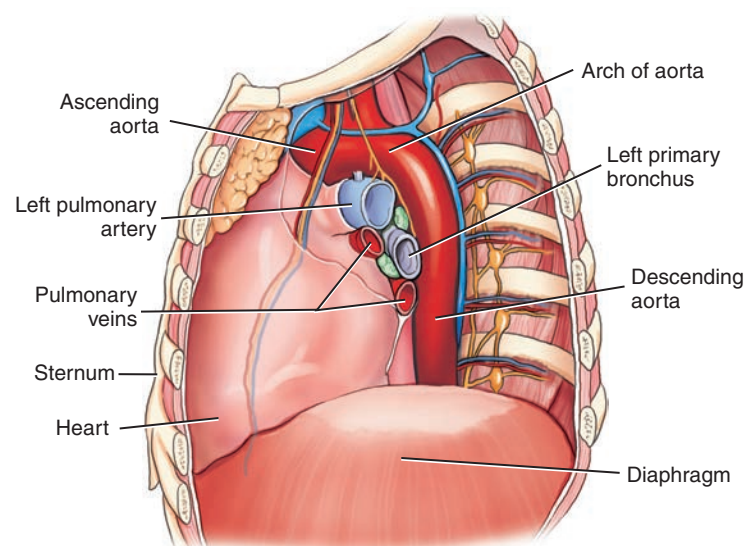


Fig. 2.24 Mediastinal relationships on left side with lung removal.

RADIOGRAPHIC POSITIONING

Body Habitus

Body habitus requires special consideration in chest radiography. There are four different styles of body build or habitus (Fig. 2.25). The four types of body habitus were explained in Chapter 1. A massively built **hypersthenic** patient has a thorax that is very **broad** and very **deep** from front to back but is **shallow** in vertical dimension, as is shown in the PA radiograph in Fig. 2.26. Therefore, care must be taken that the sides or the costophrenic angles are not cut off on a PA chest radiograph. Careful centering is also required on the lateral projection to ensure that the anterior or posterior margins are included on the radiograph.

The other extreme is a slender **asthenic** patient. With this build, the thorax is **narrow** in width and **shallow** from front to back but is very **long** in its vertical dimension. Therefore, in positioning for a chest radiograph, the technologist must ensure that the IR is positioned to include both the upper apex areas, which extend well above the clavicles, and the lower costophrenic angles. A chest PA radiograph in a nearer average **hyposthenic** patient is shown in Fig. 2.27. Care in vertical collimation for such patients must be exercised so that the costophrenic angles are not cut off on the lower margin.

Breathing Movements

Movements of the bony thorax during inspiration (taking air in) and expiration (expelling air) greatly change the dimensions of the thorax and the thoracic volume. To increase the volume of the chest during inspiration (Fig. 2.28), the thoracic cavity increases in diameter in **three dimensions**.

The first of these is the **vertical diameter**, which is increased primarily by contraction and downward movement of the diaphragm, increasing the thoracic volume.

The **transverse diameter** is the second dimension that is increased during inspiration. The ribs swing outward and upward, and this increases the transverse diameter of the thorax.

The third dimension is the **anteroposterior diameter**, which is also increased during inspiration by the raising of the ribs, especially the second through sixth ribs.

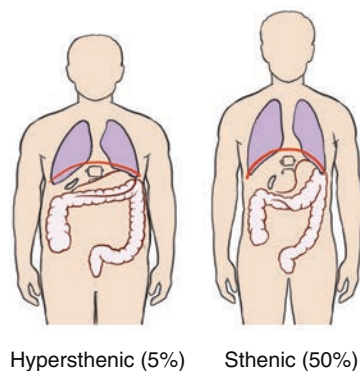
During expiration, the elastic recoil of the lungs, along with the weight of the thoracic walls, causes the three diameters of the thorax to return to normal (Fig. 2.29).

DEGREE OF INSPIRATION

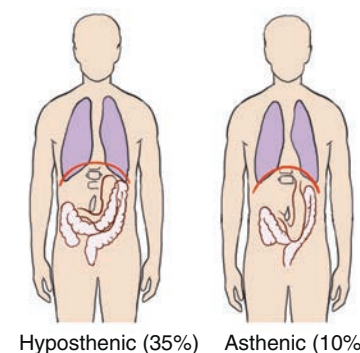
To determine the degree of inspiration in chest radiography, one should be able to identify and count the rib pairs on a chest radiograph. The first and second pairs are the most difficult to locate. When a chest radiograph is taken, the patient should take as deep a breath as possible and then hold it to aerate the lungs fully. Taking a second deep breath before holding it allows for a deeper inspiration.

The best method that can be used to determine the degree of inspiration is to observe how far down the diaphragm has moved by counting the pairs of posterior ribs in the lung area above the diaphragm. A general rule for average adult patients is to “show” a **minimum of 10** pairs of ribs on a good PA chest radiograph. To determine this, start at the top with the first rib and count down to the **tenth** or **eleventh** rib posteriorly. The posterior part of each rib, where it joins a thoracic vertebra, is the most superior part of the rib. The diaphragm should always be checked to see that it is below the level of at least the tenth posterior rib (see following NOTE). (Fig. 2.30 shows 11 posterior ribs, which can be expected in many healthy patients.)

NOTE: Patients with pulmonary diseases and trauma may be unable to inspire deeply. Therefore, it may be impossible to demonstrate 10 ribs above the diaphragm for these chest projections.



Hypersthenic (5%) Sthenic (50%)



Hyposthenic (35%) Asthenic (10%)

Fig. 2.25 Body habitus.

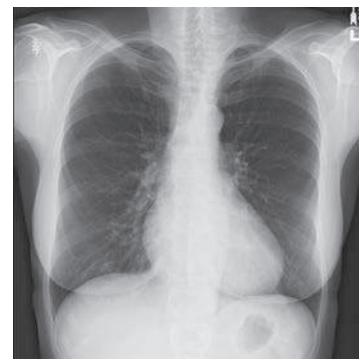


Fig. 2.26 PA (hypersthenic).

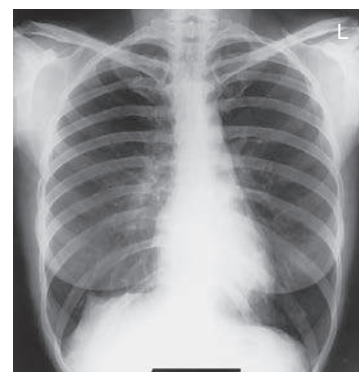


Fig. 2.27 PA (hyposthenic).

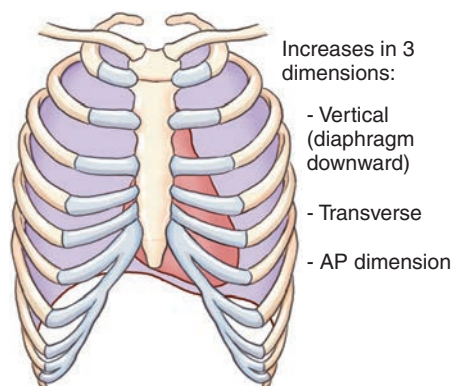


Fig. 2.28 Inspiration.

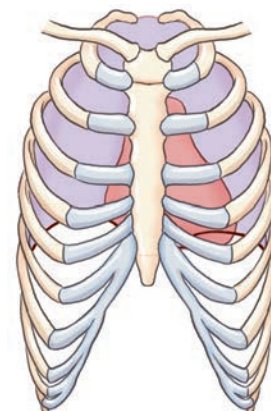


Fig. 2.29 Expiration.

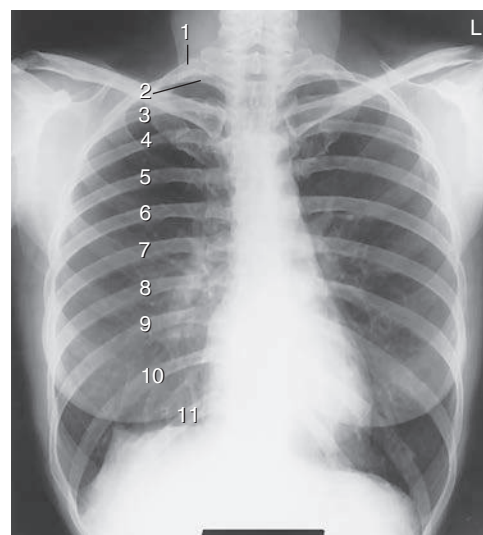


Fig. 2.30 Posterior ribs.

Positioning Considerations

Patient preparation for chest radiography includes the removal of all opaque objects from the chest and neck regions, including clothes with buttons, snaps, hooks, or any objects that would be visualized on the radiograph as a shadow (radiopaque artifact). To ensure that all opaque objects are removed from the chest region, the usual procedure is to ask the patient to remove all clothing, including bras, necklaces, or other objects around the neck. The patient then puts on a hospital gown, which commonly has the opening in the back.

Long hair may be visible as an artifact on chest radiographs taken with digital imaging systems. Long hair should be drawn up or draped across the shoulder to eliminate superimposition within the chest anatomy. Hair that is braided or tied together in bunches with rubber bands or other fasteners may also cause suspicious shadows on the radiograph if left superimposing the chest area. Oxygen lines or electrocardiogram (ECG) monitor leads should be moved carefully to the side of the chest if possible. All radiopaque objects should be moved carefully from the radiographic field of interest to prevent artifacts from interfering with the quality of the diagnostic image.

Radiation Protection

Patients should be protected from unnecessary radiation for all diagnostic radiographic examinations, especially for chest radiographs, because these are the most common of all radiographic examinations.

REPEAT EXPOSURES

Although the chest radiographic examination often is considered the simplest of all radiographic procedures, it also is the examination with the highest number of repeats in many radiology departments. Therefore, unnecessary radiation exposure from repeat exposures should be minimized by taking extra care in positioning, CR centering, and selecting correct exposure factors if automatic exposure control (AEC) systems are not used. Reduce patient dose as much as possible through the use of correct radiation protection practices by close collimation and protective shielding.

COLLIMATION

Careful collimation is important in chest radiography. Restricting the primary x-ray beam by collimation not only reduces patient dose by reducing the volume of tissue irradiated but also improves image quality by reducing scatter radiation.

LEAD SHIELDING

In addition to careful collimation, a lead shield should be used to protect the abdominal area below the lungs. This shielding is especially important for children, pregnant women, and all individuals of childbearing age. However, many departments have a general policy of shielding for all patients undergoing chest radiography.

A common type of shield for chest radiography is a type of freestanding, adjustable mobile shield placed between the patient and the x-ray tube. A vinyl-covered lead shield that ties around the waist can also be used. Both of these types of shields should provide shielding from the level of the iliac crests, or slightly higher, to the midhigh area.

BACKSCATTER PROTECTION

To protect the gonads from scatter and secondary radiation from the IR holder device and the wall behind it, some references suggest that a freestanding shield or a wraparound shield also should be placed over the radiosensitive structures outside the anatomy of interest between the patient and the IR.

TECHNICAL FACTORS

Kilovoltage

Kilovoltage (kVp) should be high enough to result in sufficient contrast to demonstrate the many shades of gray needed to visualize finer lung markings. In general, chest radiography uses **low contrast**, described as **long-scale contrast**, with more shades of gray. This requires a high kVp of 110 to 125.

Lower kVp, yielding high contrast, would not provide sufficient penetration to allow clear visualization of the fine lung markings in the areas behind the heart and lung bases. Too high contrast is evident when the heart and other mediastinal structures appear underexposed, even though the lung fields are sufficiently penetrated.

As a general rule, in chest radiography, the use of high kVp (>100) **requires the use of grids**. Moving grids or fine-line focused fixed grids can be used. Advancements in portable or mobile chest radiography include the use of low-ratio grids (6:1 or 8:1) to reduce scatter radiation and overall improve image quality.

Exposure Time and Milliamperage (mAs—Milliamperere Seconds)

Generally, chest radiography requires the use of **high mA and short exposure time** to minimize the chance of motion and resultant loss of sharpness. Sufficient mAs should be used to demonstrate the lungs and mediastinal structures without mottle or noise.

Placement of Image Markers

Throughout the positioning sections of this text, the correct or best placement of patient identification (ID) information and image markers is indicated. The top portion of each positioning page includes a drawing that demonstrates the correct IR size and placement (portrait or landscape) for computed radiography (CR) systems and the location and type of image marker used for that specific projection or position.

Although there is an assumption that the heart is located in the left thorax, there are conditions such as **situs inversus (also known as visceral inversion)**² in which the major organs of the body are on the opposite side. With this condition, the heart is located in the right thorax. An anatomic side marker (left or right) must be placed on the image receptor **prior** to exposure. If the marker is not seen radiographically, the exposure should be retaken to ensure the correct side of the thorax is identified.

PEDIATRIC APPLICATIONS

Supine Versus Erect

Generally, with newborns and small infants, for whom head support is required, chest radiographs are taken AP supine. Laterals also may be taken supine with a horizontal beam to demonstrate fluid levels (dorsal decubitus). However, erect PA and laterals are preferred whenever possible, with the use of immobilization devices such as the **Pigg-O-Stat** (Modern Way Immobilizers, Inc, Clifton, Tennessee) (described in [Chapter 16](#)).

Technical Factors

Lower kVp (70 to 85) and **less mAs** are required for pediatric patients with the **shortest exposure time possible** (to prevent motion). Higher-speed imaging systems or receptors generally are used with pediatric patients for two reasons: (1) to reduce the chance of motion and (2) to reduce the patient exposure dose (important because of the sensitivity of young tissue to radiation exposure). See [Chapter 16](#) for more detailed information on special positioning considerations required with pediatric patients.

GERIATRIC APPLICATIONS

CR Centering

Frequently, older patients have less inhalation capability with resultant smaller lung fields, and a **higher CR location** is required (CR to T6–T7).

Technical Factors

Certain pathologic conditions are more common in geriatric patients, such as **pneumonia** and **emphysema**, which may require different exposure factor adjustments, as described under Clinical Indications.

Instructions and Patient Handling

More care, time, and patience frequently are required when breathing and positioning requirements are explained to geriatric patients. Help and support provided to these patients during the positioning process are important. Arm supports for keeping the arms raised high for the lateral position are essential for many older patients.

BARIATRIC PATIENT CONSIDERATIONS

A bariatric patient may present positioning and centering challenges. Because of a larger body girth, the technologist may place the top of the IR 1 to 2 inches (2.5 to 5 cm) above the shoulder. Because the lung apices may not reach as high as perceived, center the CR and IR to the level of T7 rather than base centering on the levels of the shoulders. T7 remains your centering point for most chest projections. T7 is generally located at the level of the inferior angle of the scapula. If it cannot be located, the vertebra prominens may serve as a landmark to assist in locating the T7 level. See p. 84 for information on CR centering based on the vertebra prominens.

For the AP chest projection, the jugular notch is a palpable landmark on the bariatric patient. T7 is approximately 3 to 4 inches (8 to 10 cm) inferior to the jugular notch.

Breathing Instructions

Breathing instructions are very important in chest radiography because any chest or lung movement that occurs during the exposure results in blurring of the radiographic image. Chest radiographs must be taken on **full inspiration** to show the lungs fully expanded.

HOLD BREATH ON SECOND INSPIRATION

More air can be inhaled without too much strain on the **second** breath compared with the first. Therefore, the patient should be asked to **hold the second full inspiration** rather than the first. However, the full inspiration should not be forced to the point of strain that causes unsteadiness; this should be explained to the patient before the exposure as the patient is being positioned.

INSPIRATION AND EXPIRATION

Occasional exceptions have been noted to taking chest radiographs on full inspiration only. For certain conditions, comparison radiographs are taken on both **full inspiration** (Fig. 2.31) and **full expiration** (Fig. 2.32). Indicators for this include a possible small **pneumothorax** (air or gas in the pleural cavity), fixation or lack of normal movement of the diaphragm, the presence of a foreign body, and the need to distinguish between an opacity in the rib and one in the lung. When such comparison radiographs are taken, they should be labeled “inspiration” and “expiration.”

Note the number of ribs demonstrated above the diaphragm on the expiration projection. There are a greater number of ribs demonstrated above the diaphragm in the full inspiration radiograph. Also note the position of the two opacities in the right lung between inspiration and expiration projections. They shift position, which indicates they are within the lungs or pleura. Note also the number of ribs visible above the diaphragm, indicating the degree of inspiration (10 posterior ribs) and expiration (8 posterior ribs).

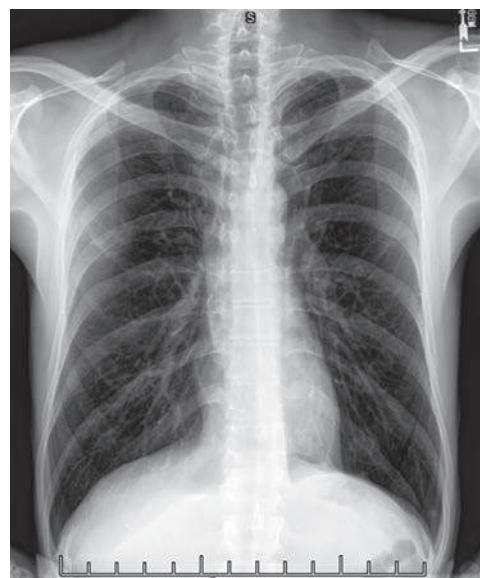


Fig. 2.31 Inspiration chest.

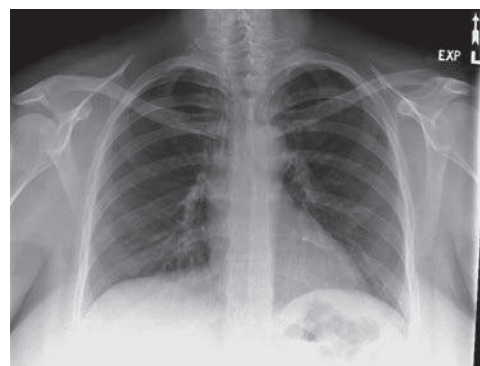


Fig. 2.32 Expiration chest.

ERECT CHEST RADIOGRAPHS

All chest radiographs should be taken in an erect position if the patient's condition allows. Three reasons for this are as follows:

1. **The diaphragm is able to move down farther.** An erect position causes the liver and other abdominal organs to drop, allowing the diaphragm to move farther down (inferior) on full inspiration and allowing the lungs to aerate fully.
2. **Air and fluid levels in the chest may be visualized.** If both air and fluid are present within a lung or within the pleural space, the heavier fluid, such as blood or pleural fluid resulting from infection or trauma, gravitates to the lowest position, whereas the air rises. In the recumbent position, a pleural effusion spreads out over the posterior surface of the lung, producing a hazy appearance of the entire lung. In the upright position, free fluid is located near the base of the lung. The PA erect chest radiograph (Fig. 2.33) shows some fluid in the left lower thoracic cavity near the base of the lung. The supine radiograph taken on a different patient (Fig. 2.34) demonstrates a generalized hazy appearance of the entire right lung, resulting from the presence of fluid now spread throughout the right thorax.
3. **Engorgement and hyperemia of pulmonary vessels may be prevented.** The term **engorgement** literally means "distended or swollen with fluid." **Hyperemia** (*hy-per-e-me-ah*) is an excess of blood that results in part from relaxation of the distal small blood vessels or arterioles.^{3,4}

An erect position tends to minimize engorgement and hyperemia of pulmonary vessels, whereas a supine position increases these, which can change the radiographic appearance of these vessels and the lungs in general.

PA 72-Inch (180-cm) Source-Image Receptor Distance

Chest radiographs taken AP rather than PA at 72 inches (180 cm) result in **increased magnification of the heart shadow**, which complicates the diagnosis of possible cardiac enlargement. The reason for this increased magnification is the **anterior location** of the heart within the mediastinum; placing it closer to the IR on the PA results in less magnification. A longer source-to-image receptor distance (SID), such as 72 inches (180 cm), magnifies less because the x-ray beam has less divergence.

Evaluation Criteria

The description for each chest projection or position in this chapter includes an evaluation criteria section. This section lists and describes specific criteria by which one can evaluate the resultant radiograph. The goal of every technologist should be to take the optimal radiograph. These criteria provide a **definable standard** by which every chest radiographic image can be evaluated to determine where improvements can be made.

Important evaluation criteria for all routine PA and lateral chest radiographs are described in the following sections.

PA CHEST POSITIONING

True PA, No Rotation

Even a slight amount of rotation on a PA chest projection results in distortion of the size and shape of the heart shadow because the heart is located anteriorly in the thorax. Therefore, it is important that there be **no rotation** (Fig. 2.35). To prevent rotation, ensure that the patient is standing evenly on both feet with both shoulders rolled forward and downward. Also, check the posterior aspect of the shoulders and the lower posterior rib cage and the pelvis to ensure no rotation. **Scoliosis** and excessive **kyphosis** make it more difficult to prevent rotation. Scoliosis is lateral, or side-to-side, curvature of the spine, which frequently is combined with excessive kyphosis, a humpback curvature. Together, these

spinal curvatures frequently result in a "twisting" deformity of the bony thorax, making a true PA without some rotation more difficult or impossible.

Rotation on PA chest radiographs can be determined by examination of both sternal ends of the clavicles for a symmetric appearance in relationship to the spine. On a true PA chest without rotation, **both the right and the left sternal ends of the clavicles are the same distance from the center line of the spine**. Note the rotation evident in Fig. 2.36 by the difference in distance between the center of the spinal column and the sternal end of the right clavicle compared with the left.

The direction of rotation can be determined by noting which sternal end of the clavicle is closest to the spine. For example, in Fig. 2.36, the left side of the thorax is rotated toward the IR (right side moved away from IR), which creates a slight left anterior oblique (LAO) that decreases the distance of the left clavicle from the spine.

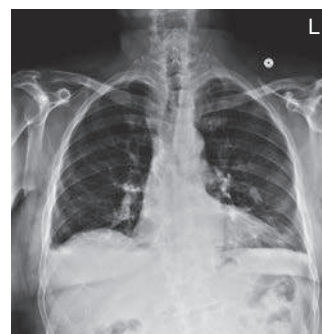


Fig. 2.33 PA erect, some fluid evident in left lower lung. (Note flat line appearance near left hemidiaphragm.)



Fig. 2.34 Supine AP chest (fluid in right lung).



Fig. 2.35 Without rotation.

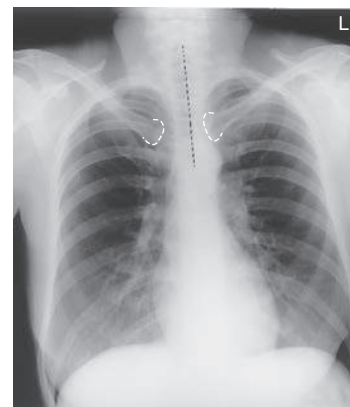


Fig. 2.36 With rotation (slight LAO).

Extending the Chin

Sufficient extension of the patient's neck ensures that the chin and neck are not superimposing the uppermost lung regions, the apices of the lungs. This is demonstrated by the two radiographs in Figs. 2.37 and 2.38. Also, ensure that the upper collimation border is high enough so that the apices are not cut off.

Minimizing Breast Shadows

A patient with large pendulous breasts should be asked to lift them up and outward and then to remove her hands as she leans against the chest board (IR) to keep them in this position. This position lessens the effect of breast shadows over the lower lung fields. However, depending on the size and density of the breasts, breast shadows over the lower lateral lung fields cannot be totally eliminated (Fig. 2.39).

LATERAL CHEST POSITIONINGWW**Side Closest to IR**

The patient's side closest to the IR is best demonstrated on the finished radiograph. A **left lateral** (Fig. 2.40) should be performed unless departmental protocol indicates otherwise or unless certain pathology in the right lung indicates the need for a right lateral. A left lateral more accurately demonstrates the heart region (without as much magnification) because the heart is located primarily in the left thoracic cavity.

True Lateral, No Rotation or Tilt

Ensure the patient is standing straight with weight evenly distributed on both feet and arms raised. As a check against rotation, confirm that the posterior surfaces of the shoulder and the pelvis are directly superimposed and perpendicular to the IR. Because of the divergent x-ray beam, the posterior ribs on the side farthest away from the IR are magnified slightly and projected slightly posterior compared with the side closest to the IR on a true lateral chest; this is more noticeable on a broad-shouldered patient. However, this separation of posterior ribs resulting from divergence of the x-ray beam at the commonly used 72-inch (180-cm) SID **should be only ¼ to ½ inch (about 1 cm)**. Any greater separation than this indicates rotation of the thorax from a true lateral position.⁴

NOTE: Some references recommend an intentional slight anterior rotation of the side away from the IR so that the posterior ribs are directly superimposed. This rotation may be preferred in some departments, but because the heart and most lung structures are near-midline structures and are not affected by the beam divergence, a straight lateral with respect to the IR is more common; this causes slight separation of the posterior ribs and costophrenic angles, as described earlier.

Fig. 2.41 shows a lateral chest with **excessive rotation**, as indicated by the **amount of separation of the right and left posterior ribs** and **separation of the two costophrenic angles**. This represents a positioning error and generally would require a repeat radiograph.

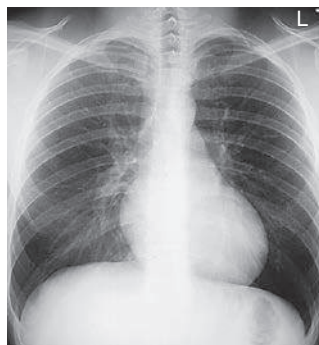


Fig. 2.37 Chin up.

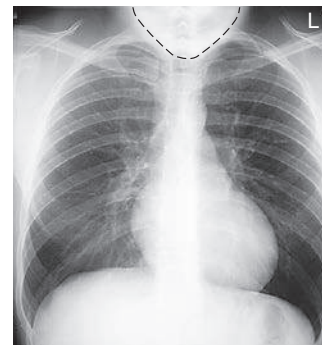


Fig. 2.38 Chin down.



Fig. 2.39 Breast shadows evident—patient has pneumonia.



Fig. 2.40 Without excessive rotation (ribs superimposed).



Fig. 2.41 Excessive rotation—positioning error (ribs not superimposed).

Direction of Rotation

The direction of rotation on a lateral chest is sometimes difficult to determine. Frequently, however, this can be done by identifying the left hemidiaphragm by the gastric air bubble in the stomach or by the inferior border of the heart shadow, both of which are associated with the left hemidiaphragm.³

No Tilt

There also should be **no tilt**, or leaning “sideways.” The **midsagittal plane must be parallel to the IR**. If the patient’s shoulders are placed firmly against the chest board (IR) on a lateral chest, the lower lateral thorax or hips or both may be 1 or 2 inches (2.5 to 5 cm) away. This is especially true on broad-shouldered patients. Tilt, if present, may be evident by closed disk spaces in the thoracic vertebra.

Arms Raised High

Ensure the patient raises both arms sufficiently high to prevent superimposition on the upper chest field. Patients who are weak or unstable may need to grasp a support (Fig. 2.42).

When the patient’s arms are not raised sufficiently, the soft tissues of the upper arm superimpose portions of the lung field, as is demonstrated in Fig. 2.43. The arrows in this figure show margins of soft tissues of the arms overlying upper lung fields. This would require a repeat and should be avoided.

CR Location

The top of the shoulder traditionally has been used for chest positioning. This method includes placing the top of the IR 1½ to 2 inches (4 to 5 cm) above the shoulders and centering the CR to the center of the IR. However, this positioning method is inconsistent, given variations in lung field dimensions owing to differences in body habitus, as demonstrated by a comparison of Figs. 2.44 and 2.45. The small circle indicates where the CR was placed on these two patients. The center of the lungs (indicated by X) is shown to be near the center of the IR for the male patient in Fig. 2.44 but is above center on the small and older female patient in Fig. 2.45. This centering error unnecessarily exposes a large portion of the upper abdomen.

These variations demonstrate the importance of a chest positioning method that consistently centers the CR to the center of the lung fields on all types of patients with accurate collimation on *both* top and bottom.

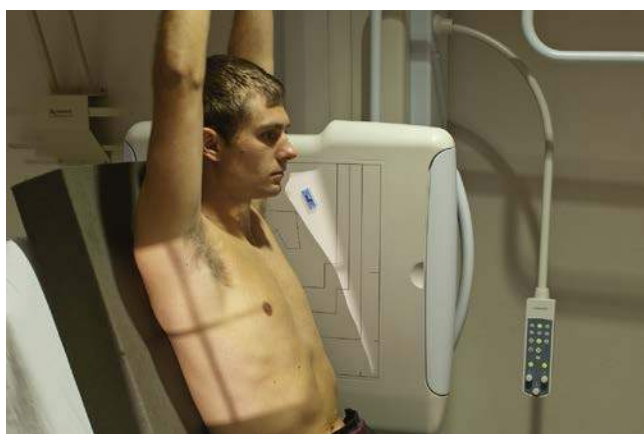


Fig. 2.42 Arms raised high.

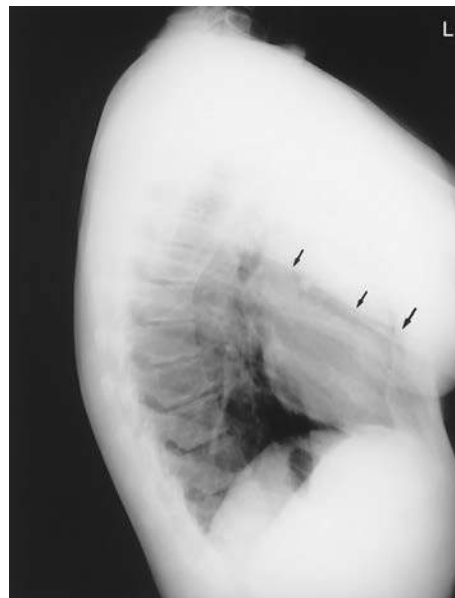


Fig. 2.43 Arms not raised—positioning error.

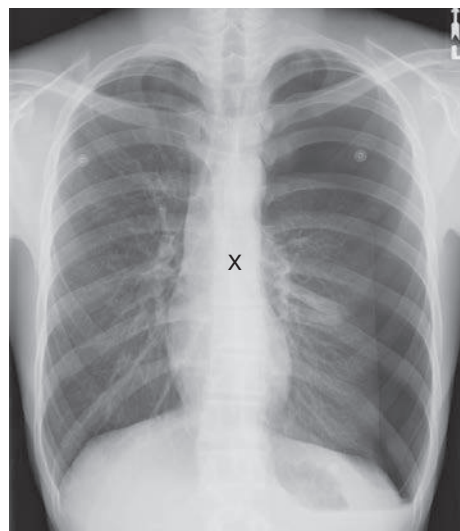


Fig. 2.44 Average sthenic/hyposthenic male patient (correct CR and collimation).

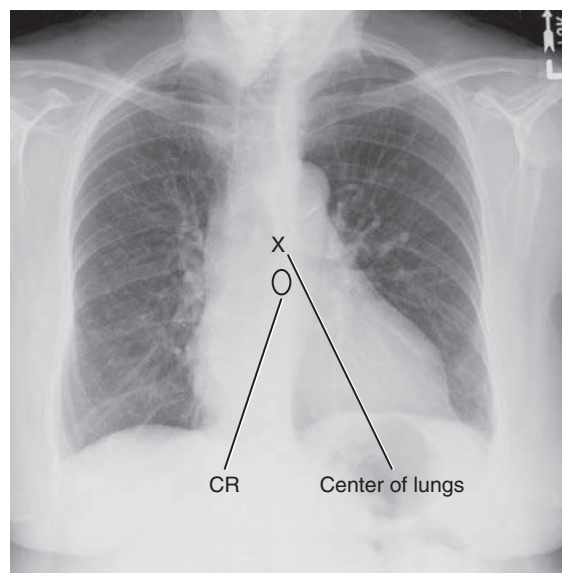


Fig. 2.45 Small and older female patient (incorrect CR and collimation and tilt).

CR CHEST-POSITIONING METHOD

Bony landmarks are consistent and reliable as a means of determining CR locations. Landmarks for locating the center of the lung fields are as follows.

Vertebra Prominens (PA Chest)

The vertebra prominens corresponds to the level of T1 and the uppermost margin of the apex of the lungs. This landmark, which can be palpated at the base of the neck, is the preferred landmark for locating the CR on a PA chest (Figs. 2.46 and 2.47). For an average adult female patient, this is down about 7 inches (18 cm); for an average adult male patient, this is down about 8 inches (20 cm).

One method of determining this distance is by using an average hand spread, as shown in Fig. 2.48. Most hands can reach 7 inches (18 cm). The 8-inch (20-cm) distance can be determined by estimating an additional inch. If the hand spread method is used, the technologist should practice with a ruler to determine these distances consistently.

These differences between male and female are true for near-average body types in the general population, with crossover exceptions in which certain larger athletic women may have longer lung fields and some men may have shorter lungs. However, for purposes of chest positioning for the general population, the average measurements of **7 inches (18 cm) for a woman** and **8 inches (20 cm) for a man** can be used as reliable guidelines (Fig. 2.49).

Exceptions

Other noteworthy exceptions in centering involve variations in body type. For example, the author found that well-developed athletic sthenic/hyposthenic body types require centering nearer to T8, or 9 inches (23 cm) down from the vertebra prominens. The hypersthenic body type requires centering only from 6 to 7 inches (15 to 18 cm) down.

NOTE: For most patients, this CR level for PA chests is near the level of the inferior angle of the scapula, which corresponds to the level of T7 on an average patient.

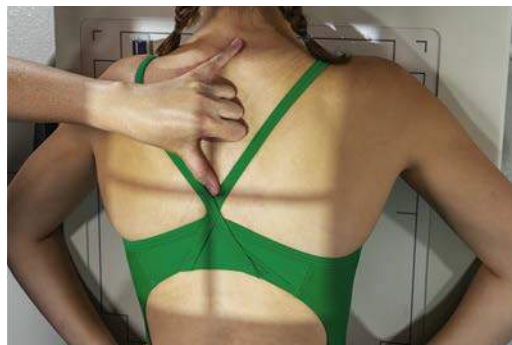


Fig. 2.47 Correct CR using vertebra prominens. Distance on an average female is 7 inches (18 cm).



Fig. 2.48 Hand spread method—7 to 8 inches (18 to 20 cm).



Fig. 2.46 Correct CR using vertebra prominens. Distance on an average male is 8 inches (20 cm).

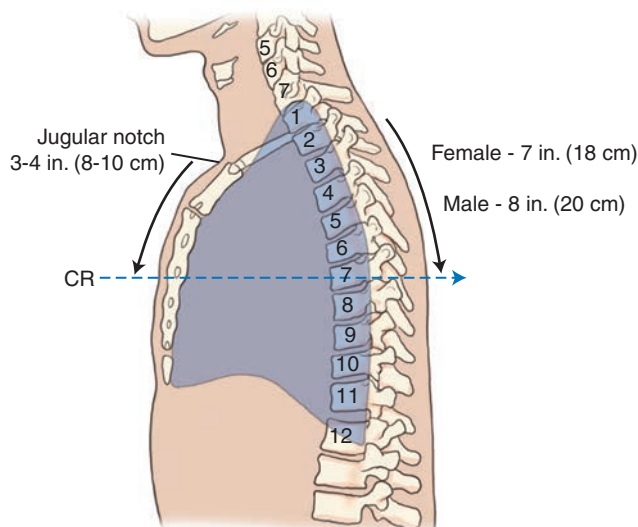


Fig. 2.49 Topographic landmark for AP chest. CR, Central ray.

Jugular Notch (AP Chest)

The easily palpated jugular notch is the recommended landmark for location of the CR for AP chest radiographs. The level of T7 on an average adult is 3 to 4 inches (8 to 10 cm) below the jugular notch. For most older or hypersthenic patients, this is approximately **3 inches (8 cm)**. For younger or sthenic/hyposthenic athletic types, this is nearer **4 to 5 inches (10 to 13 cm)**.

This distance can be determined by the technologist's hand width. The average-sized hand width with the fingers together is approximately 3 inches (8 cm) as shown in Fig. 2.50.

Lung Dimensions and IR Placement

Contrary to common belief, the **width or horizontal dimension of the average PA or AP chest is greater than the vertical dimension**. If using computed radiography or analog systems, the technologist should use his or her discretion to determine whether the IR should be placed portrait (lengthwise) or landscape (crosswise) for PA or AP projections based on the size and body habitus of the patient, ensuring that the right and left costophrenic angles of the lungs are imaged.

For recumbent AP chest radiographs (usually taken at <72 inches [180 cm], with an accompanying increase in divergence of the x-ray beam), the chance that the side borders of the lungs may be cut off is increased when the IR is placed portrait. It is recommended that for most AP chest radiographs, the 14 × 17-inch (35 × 43-cm) IR should be placed landscape. The IR and CR should be centered to a point 3 to 4 inches (8 to 10 cm) below the jugular notch (see Fig. 2.50).

PA Chest

Some erect PA chests are performed with direct digital chest systems, which have a 17 × 17-inch (43 × 43-cm) IR that will accommodate both long and broad chest dimensions. Proper collimation with these systems is imperative. Ideally, the field is collimated on all four sides to the area of the lung fields. At a minimum, the field should be collimated to 14 × 17 inches (35 × 43 cm) or smaller. As the patient is standing and facing the chest IR, one can determine the amount of collimation by standing behind the patient and placing one's hands squarely on each side of the chest. If there is any doubt that both sides of the chest can be included, the IR should be collimated in a landscape direction, because the height of the average lung field is less than the width.

Collimation Guidelines

Side collimation borders can be determined easily by adjusting the illuminated field margins to the **outer skin margins** on each side of the posterior chest surface (given that lungs expand during deep inspiration). However, the upper and lower collimation borders are more difficult to determine, because these lung margins are not visible externally.

A reliable method for upper and lower chest collimation is to adjust the upper border of the illuminated light field to the **vertebra prominens**, which (with the divergent rays) results

in an upper collimation margin on the IR of about 1½ inches (4 cm) above the vertebra prominens (Fig. 2.51). This also results in a lower collimation border of 1 to 2 inches (2.5 to 5 cm) below the costophrenic angles if the CR was centered correctly. These distances above and below the lungs allow for some margin of error in CR placement without cutting off upper or lower lungs.



Fig. 2.50 IR landscape, CR 3 to 4 inches (8 to 11 cm) below jugular notch.



Fig. 2.51 Collimation guidelines, PA chest: CR—T7 or T8; sides—outer skin margins; upper—level of vertebra prominens.

Digital Imaging Considerations

Guidelines as listed next should be followed when chest images are acquired through the use of digital imaging technology. (See Chapter 1 for a discussion of applications of digital technology.)

1. **Collimation.** In addition to the benefit of reducing radiation dose to the patient, collimation that is closely restricted to the part that is being examined is key to ensuring the image processed by the computer is of optimal quality. Close collimation also improves image quality by preventing secondary and scatter radiation from surrounding areas (e.g., the dense abdomen below) from reaching the highly sensitive photostimulable storage phosphor plate (PSP) or digital radiography IR. With the digital IR being larger than other IRs, close collimation is critical for patient dose reduction and improved image quality. Close collimation also allows the computer to provide accurate information regarding the exposure indicator.
2. **Accurate centering.** Because of the exposure factors used for the digital image receptor, it is important that the body part and CR be accurately centered to the IR. In chest imaging, this involves centering the CR to the center of the lung fields, as described previously.
3. **Exposure factors.** Digital imaging systems are known for their wide exposure latitude; they are able to process an acceptable image from a broad range of exposure factors (kVp and mAs). However, the ALARA (as low as reasonably achievable) principle related to patient exposure still must be followed, and the lowest exposure factors required to obtain a diagnostic image must be used. This includes using the highest possible kVp and the lowest mAs consistent with optimal image quality.
4. **Post-processing evaluation of exposure indicator.** When the image is available for viewing, it is critiqued for positioning and exposure accuracy. The technologist must also **check the exposure indicator** to verify that the exposure factors used were in the correct range to ensure optimal quality with the least radiation to the patient. A standardized exposure index (EI) has been established for new digital systems; however, CR and digital systems without recent upgrades may have varying acceptable exposure ranges. It is important to learn the acceptable ranges for the exposure indicator for each digital system.

Alternative Modalities and Procedures

Conventional Tomography and CT

CT is performed frequently to examine and identify pathology of the mediastinum and lungs. Multi-detector computed tomography (MDCT) or **multislice CT** (MSCT) provides for faster scanning due to its ability to acquire numerous slices within one rotation of the gantry, which is especially advantageous in the thoracic region. Multidetector CT scanners can produce high-quality images of the heart and lungs with just one breath hold required of patients. MDCT is now routinely used in cardiac angiography for the demonstration of calcification and/or stenosis of the coronary arteries (see Chapter 18).

Sonography

Sonography (ultrasound) may be used to detect **pleural effusion** (fluid within pleural space) or for guidance when a needle is inserted to aspirate the fluid (thoracentesis).

An **echocardiogram** is an ultrasound examination in which sound waves are used to create an image of the heart. (This is not the same as an electrocardiogram [ECG], which is a completely different type of examination that assesses the electrical activity of the heart.) This imaging modality is valuable to demonstrate the dynamic movement and function of the heart valves.

Nuclear Medicine

Certain nuclear medicine procedures involving radionuclides can be used to evaluate and diagnose pulmonary diffusion conditions

or pulmonary emboli. With the use of single-photon emission computed tomography (SPECT), the heart can be evaluated specifically for myocardial infarction.

MRI

Cardiovascular MRI procedures can be performed to demonstrate and evaluate pathology including congenital heart disorders, graft patency, cardiac tumors, thrombi, pericardial masses, and aortic dissection and aneurysm. MRI is unlikely to replace echocardiography for cardiac evaluation. However, MRI is a viable supplement to other imaging modalities, because it can provide multiplanar views of tumors and masses, further assess mediastinal pathology, and evaluate aortic dissection and aneurysm.

Clinical Indications

The **clinical indications** (Table 2.1), as listed subsequently and in each chapter of this textbook, are not intended to be inclusive of all diseases or pathologic conditions of which technologists should be aware or that may be covered in a separate pathology course. However, they do represent conditions that are encountered more commonly, and knowledge and understanding of these clinical indications should be considered **routine** and **essential** for all technologists.

Patient histories in which these clinical indications are noted help the technologist select the optimum exposure factors and ensure that the necessary projections or body positions are being used. When adjusting the exposure factors, the technologist must ensure that a quality diagnostic image is obtained without obscuring or accentuating the disease process.

This information is also important for the technologist in understanding and being prepared to respond to patient needs and reactions during the radiographic procedure. For the chest, these clinical indications are numerous and complex. The more common indications for youth and adults are listed as follows (see Chapter 16 for information on infants and children).

CHEST PATHOLOGY, CONDITIONS, AND TRAUMA⁵

Aspiration (*as-pi-ra'-shun*) (mechanical obstruction) is most common in small children when **foreign objects** are swallowed or aspirated into the air passages of the bronchial tree. In adults, it may occur with food particles, creating coughing and gagging (relieved by the Heimlich maneuver). Aspiration may be evident in the lower airways on frontal and lateral chest radiographs or AP and lateral radiographs of the upper airway.

Atelectasis (*at'-e-lek-tah-sis*) is a condition rather than a disease, in which collapse of all or a portion of a lung occurs as the result of obstruction of the bronchus or puncture or "blowout" of an air passageway. With less air in the lung than normal, this region appears more radiodense, and this may cause the trachea and heart to shift to the affected side.

Bronchiectasis (*brong'-ke-ek-tah-sis*) is an irreversible dilation or widening of bronchi or bronchioles that may result from repeated pulmonary infection or obstruction. Areas of bronchial walls are destroyed and become chronically inflamed, resulting in increased production of mucus and causing chronic cough and expectoration (coughing up sputum). Pus can collect in dilated regions, resulting in an increase in regional radiodensity with less air in these regions (most common in the lower lobes).

Bronchitis (*brong-ki-tis*) is an acute (short-term) or chronic (long-term) condition in which excessive mucus is secreted into the bronchi, causing cough and shortness of breath. The chief cause is cigarette smoking. Infectious bronchitis is caused by viruses or bacteria. Bronchitis generally involves lower lobes and in severe cases is demonstrated on radiographs by hyperinflation and more dominant lung markings.

TABLE 2.1 SUMMARY OF CLINICAL INDICATIONS

CONDITION OR DISEASE	MOST COMMON RADIOGRAPHIC EXAMINATION	POSSIBLE RADIOGRAPHIC APPEARANCE	EXPOSURE FACTOR ADJUSTMENT ^a
Aspiration (mechanical obstruction)	PA and lateral chest and lateral upper airway	Radiodense or radiopaque outline	Soft tissue technique for upper airway (–)
Atelectasis (collapse of all or portion of lung)	PA and lateral chest and PA inspiration/expiration	Radiodense lung regions with shift of heart and trachea in severe cases	Increase (+)
Bronchiectasis	PA and lateral chest with bronchogram or CT	Radiodense lower lungs	Generally none
Bronchitis	PA and lateral chest	Hyperinflation (general radiolucency) and dominant lung markings of lower lungs	Generally none
Chronic obstructive pulmonary disease (COPD)	PA and lateral chest	Depends on underlying cause	Changes in severe cases only
Cystic fibrosis	PA and lateral chest	Increased radiodensities in specific lung regions	Increase with severe condition (+)
Dyspnea (difficult breathing)	PA and lateral chest	Depends on cause of dyspnea	Depends on cause
Emphysema	PA and lateral chest	Increased lung dimensions, barrel chest, flattened diaphragm, radiolucent lungs	Significantly decreased, dependent on severity (–)
Epiglottitis	Soft tissue lateral upper airway	Narrowing of upper airway at epiglottic region	Soft tissue lateral technique (–)
Lung neoplasm			
Benign (hamartoma)	PA and lateral chest	Radiodensities with sharp outlines; mass may be calcified (radiopaque)	Generally none
Malignant types	PA and lateral chest, CT scans	Slight shadows in early stages, larger defined radiopaque masses in advanced stages	Generally none
Pleural effusion (hydrothorax) (in pleural cavity)	Erect PA and lateral chest or horizontal beam lateral decubitus with affected side down	Increased radiodensity, air-fluid levels, possible mediastinal shift (see Atelectasis)	Increase (+)
Empyema (fluid is pus)			
Hemothorax (fluid is blood)			
Pleurisy	PA and lateral chest	Possible air-fluid levels, or none with “dry” pleurisy	Generally none
Pneumonia (pneumonitis)	PA and lateral chest	Patchy infiltrate with increased radiodensity	Generally none
Aspiration pneumonia			
Bronchopneumonia			
Lobar (pneumococcal)			
Viral (interstitial)			
Pneumothorax	Erect PA and lateral chest or lateral decubitus with affected side up , PA inspiration/expiration for small pneumothorax	Lung seen displaced from chest wall, no lung markings	Generally none
Pulmonary edema (fluid within lungs)	PA and lateral chest; horizontal beam projection for air-fluid levels	Increased diffuse radiodensity in hilar regions; air-fluid levels	Increase (+) in severe cases
Pulmonary emboli (sudden blockage of artery in lung)	PA and lateral chest and perfusion scans (nuclear medicine), CT scans	Rarely demonstrated on chest radiographs except for possible wedge-shaped opacity (Hampton’s hump)	Generally none
Respiratory distress syndrome (RDS)—commonly called hyaline membrane disease (HMD) in children	PA and lateral erect chest	Granular pattern of increased radiodensity throughout lungs, possible air-fluid levels	Increase (+) without obscuring pathology
Tuberculosis			
Primary tuberculosis	PA and lateral chest	Small opaque spots throughout lungs; enlargement of hilar region in early stages	Generally none
Reactivation (secondary) tuberculosis	PA and lateral chest and AP lordotic chest, tomograms	Regions of calcification with cavitations, frequently in area of upper lobes and apices with upward retraction of hila	None or increase slightly (+)
Occupational lung diseases (forms of pneumoconiosis)			
Anthraxis (black lung)	PA and lateral chest	Small opaque spots throughout lungs	Generally none
Asbestosis	PA and lateral chest	Calcifications (radiodensities) involving the pleura	Generally none
Silicosis	PA and lateral chest	Distinctive pattern of scarring and dense nodules	Generally none

^aAutomatic exposure control (AEC) systems are designed to optimize mAs. Digital radiographic systems will correct exposure brightness automatically for patient size variances and for these pathologic conditions through processing algorithms; manual adjustments generally are not needed when AEC is used if the AEC system is calibrated correctly and used as intended. However, these exposure adjustments may be needed for more extreme cases or for repeats, even with AEC. Manual exposure adjustments are also important when manual exposure techniques such as for tabletop or mobile examinations are set when AEC is not used.

Chronic obstructive pulmonary disease (COPD) is a form of persistent obstruction of the airways that usually causes difficulty in emptying the lungs of air; it may be caused by emphysema or chronic bronchitis (smoking is the predominant cause of COPD). Asthma also is considered a COPD. Mild cases of COPD usually are not detectable on chest radiographs, but more severe conditions are clearly demonstrated (see **emphysema** later in this list).

Cystic fibrosis (*sis'-tik fi-bro'-sis*), the most common of inherited diseases, is a condition in which secretions of heavy mucus cause progressive “clogging” of bronchi and bronchioles. This may be evident on chest radiographs as increased radiodensities in specific lung regions, along with hyperinflation.

Dyspnea (*disp'-ne-ah*) is a condition of shortness of breath, which creates a sensation of difficulty in breathing; it is most common in older persons. Although generally caused by physical exertion, it may be caused by restrictive or obstructive defects within the lungs or airways. Dyspnea also may be caused by pulmonary edema related to cardiac conditions. PA and lateral chest radiographs are commonly taken as an initial procedure followed by other examinations in an effort to make a diagnosis.

Emphysema (*em'-fi-se-mah*) is an irreversible and chronic lung disease in which air spaces in the alveoli become greatly enlarged as a result of alveolar wall destruction and loss of alveolar elasticity. Air tends not to be expelled during expiration, resulting in seriously labored breathing with impedance of gas exchange within the lungs. Causes include smoking and long-term dust inhalation. In severe cases, emphysema is evident on chest radiographs by **increased lung dimensions**, barrel chest with depressed and flattened diaphragm obscuring costophrenic angles, and an elongated heart shadow. Lung fields appear very **radiolucent**, requiring a significant **decrease** in exposure factors from a normal chest, even with the increased chest dimensions.

Epiglottitis (*ep'-i-glo-ti'-tis*) is most common in children ages 2 to 5. See [Chapter 16](#) for more information on this serious, **life-threatening condition, which can develop very rapidly**. A soft tissue lateral of the upper airway may demonstrate edema or swelling at the point of the epiglottis.

Lung neoplasm refers to a new growth or tumor. Neoplasms may be benign (noncancerous) or malignant (cancerous).

- **Benign:** A **hamartoma** (*ham'-ahr-to'-ma*) is the most common benign pulmonary mass, and it generally is found in peripheral regions of the lungs. These are seen on chest radiographs as small radiodense masses with sharp outlines.
- **Malignant:** Many types of lung cancers have been identified, and more than 90% start in the bronchi (bronchogenic carcinoma). Less common is alveolar cell carcinoma, which originates in the alveoli of the lungs. Also, many cancers, such as breast, colon, and prostate, start elsewhere in the body before spreading to the lungs as **pulmonary metastases**. Studies have shown that smoking is the primary cause in about 90% of all lung cancers in men. In the past 50 years, a woman's risk of dying from lung cancer from smoking has more than tripled and is nearly equal to men's risk.⁶

Lung cancer may be demonstrated on chest radiography as slight shadows in the early stages and as more sharply defined, larger radiopaque masses in more advanced cases. Malignant lung tumors rarely calcify; therefore, calcified radiopaque masses or nodules are generally benign.

CT scans may reveal small nodules that are not yet seen on chest radiographs. Biopsies usually are required to determine whether these shadows are the result of inflammation or are cancerous.

Occupational lung disease (forms of pneumoconiosis) arises from occupational exposures, including certain types of mine work, sandblasting, and similar professions. Chest x-rays show distinctive patterns of nodules and scarring densities.

- **Anthracois** (*an'-thre-ko'-sis*), also called **black lung pneumoconiosis**, is caused by deposits of coal dust. With long-term inhalation (≥ 10 years), it spreads throughout the lungs and is seen on chest radiographs as small opaque spots or conglomerate masses.
- **Asbestosis** (*as'-bes-to'-sis*) is caused by inhalation of asbestos dust (fibers), which results in pulmonary fibrosis. It may develop into lung cancer, especially in smokers.
- **Silicosis** (*sil'-i-ko'-sis*) is a permanent condition of the lungs that is caused by inhalation of silica (quartz) dust, a form of sand dust. Patients with silicosis are three times more likely to develop TB than are persons without silicosis.⁷

Pleural effusion (an older, outdated term is hydrothorax) is a condition of abnormal accumulation of fluid in the pleural cavity. Types of pleural effusion include the following:

- **Empyema** (*em'-pi-e'-mah*), which occurs when the fluid is pus. Empyema may be caused by chest wounds, obstruction of bronchi, or ruptured lung abscess. It may develop when pneumonia or a lung abscess spreads into the pleural space.
- **Hemothorax** (*he'-mo-thor-aks*), which occurs when the fluid is blood. A common cause of right-sided or bilateral pleural effusion is congestive heart failure. Causes of left-sided effusion include trauma, pulmonary infarct, pancreatitis, and subphrenic abscess.

Any type of pleural effusion is demonstrated by **fluid levels** on horizontal-beam chest radiographs. Small amounts are best shown by a lateral decubitus position with **affected side down** or with **erect positioning**.

Pleurisy (*ploor'-i-se*) is characterized by inflammation (usually caused by a virus or bacterium) of the pleura surrounding the lungs. The cause is visceral and parietal pleura “rubbing” during respiration, which results in severe pain. It frequently follows pneumonia or trauma to the chest. Pleurisy may be demonstrated radiographically by associated pleural effusion. A condition called “dry pleurisy” does not include fluid accumulation and generally is not visible on radiographs.

Pneumonia (*noo-mon'-ya*) (pneumonitis) is an inflammation of the lungs that results in **accumulation of fluid** within certain sections of the lungs, creating increased radiodensities in these regions. The most common initial diagnostic examination consists of PA and lateral erect horizontal beam radiographs. Types of pneumonia are derived from the location and cause of the inflammation. Normal exposure factors generally are used initially. The radiologist may request secondary images with increased density (brightness) to see through the area of interest to rule out a lesion in the same anatomic region when film-screen imaging methods are being used. The different types of pneumonia include the following:

- **Aspiration pneumonia** is caused by aspiration of a foreign object or food into the lungs, which irritates the bronchi, resulting in edema.
- **Bronchopneumonia** is bronchitis of both lungs that most commonly is caused by *Streptococcus* or *Staphylococcus* bacteria.
- **Lobar pneumonia** generally is confined to one or two lobes of the lungs.
- **Viral (interstitial) pneumonia** causes inflammation of the alveoli and connecting lung structures. It most commonly is evident as increased radiodensities in the region surrounding the hila.

Pneumothorax (*noo-mo-thor-aks*) is an accumulation of air in the pleural space that causes partial or complete collapse of the lung and results in immediate and severe shortness of breath and chest pain. It may be caused by trauma or a pathologic condition that causes spontaneous rupture of a weakened area of lung.

Radiographically, the affected lung can be seen displaced away from the chest wall. Most evident on chest radiographs is the fact that **no lung markings** are seen in the region of the collapsed lung. Care should be taken to identify the lung edge or boundary. Chest radiographs for pneumothorax should be taken **erect**. If the patient cannot assume an erect position, a horizontal beam **lateral decubitus** position with the **affected side up** should be taken (not down as with pleural effusion).

Erect PA inspiration/expiration radiographs often are taken to demonstrate a small pneumothorax, which is best seen at the apex of an erect PA radiograph with maximum expiration.

Pulmonary edema is a condition of excess fluid within the lung that most frequently is caused by a backup in pulmonary circulation commonly associated with congestive heart failure. A common cause is coronary artery disease, in which blood flow to the heart muscle is restricted. Coronary artery disease weakens the heart and results in inadequate pulmonary circulation, causing backup of blood in the lungs. The condition is seen on chest radiographs as a diffuse increase in radiodensity in the hilar regions fading toward the periphery of the lung and as increased air-fluid levels with horizontal beam projections in more severe conditions.

Respiratory distress syndrome (RDS) (commonly called **hyaline membrane disease [HMD]** in infants and **adult respiratory distress syndrome [ARDS]** in adults) is an emergent condition in which the alveoli and capillaries of the lung are injured or infected, resulting in leakage of fluid and blood into the spaces between alveoli or into the alveoli themselves with formation of hyaline membranes. (HMD results from a lack of lung development in which the alveoli collapse as the result of lack of internal tension.) This leakage can be detected radiographically as

increased density (brightness) throughout the lungs in a granular pattern as the normally air-filled spaces are filled with fluid. The most common radiographic sign is an "air bronchogram."

Tuberculosis (*too-ber-ku-lo-sis*) (TB) is a contagious disease (potentially fatal) that is caused by airborne bacteria. At one time, TB resulted in more than 30% of all deaths, but the development of vaccines and antibiotics such as streptomycin in the 1940s and 1950s nearly eliminated the threat of this disease. However, occurrence of TB has begun to increase again with the increased incidence of acquired immunodeficiency syndrome (AIDS) and in the presence of urban overcrowding and unsanitary conditions.

- **Primary tuberculosis** refers to TB that occurs in persons who have never had the disease before. Hilar enlargement, along with enlarged mediastinal lymph nodes, is an important indicator of primary TB. Small focal spot lesions may be found anywhere in the lungs, and unilateral pleural effusion is common, especially in adults.
- **Reactivation (secondary) tuberculosis** usually develops in adults and generally is first evident on radiography bilaterally in the upper lobes as irregular calcifications that are mottled in appearance. Upward retraction of the hila is frequently evident. As healing occurs, fibrous tissue develops, with calcification surrounding the region and leaving a type of cavity that can be seen on tomograms of this region. **AP lordotic projections** are frequently requested for visualization of calcifications and cavitations of the apices and upper lobes.

Routine and Special Projections

The chest projections shown and described on the following pages are suggested routine and special departmental projections that all student technologists should master.

Routine projections are those that are commonly taken on average patients who are helpful and can cooperate in performing the procedure.

Special projections are more common projections that are taken as extra or additional projections to demonstrate better certain pathologic conditions or specific body parts or when the patient is unable to cooperate fully.

PA PROJECTION—CHEST

AMBULATORY PATIENT

Clinical Indications

- When performed erect, PA demonstrates pleural effusion, pneumothorax, atelectasis, and signs of infection

Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait or landscape (see NOTE 1)
- Grid
- kVp range—110–125

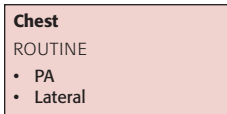
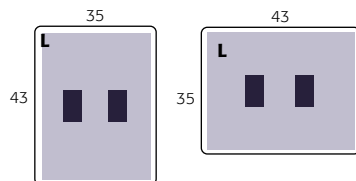


Fig. 2.52 PA chest.



Fig. 2.53 PA chest.

Shielding Shield radiosensitive tissues outside region of interest

Patient Position

- Patient erect, feet spread slightly, weight equally distributed on both feet
- Chin raised, resting against IR
- Hands on lower hips, palms out, elbows partially flexed (Fig. 2.52)
- Shoulders rotated forward against IR to allow scapulae to move laterally clear of lung fields; shoulders depressed downward to move clavicles below the apices

Part Position

- Align midsagittal plane with CR and with midline of IR with equal margins between lateral thorax and sides of IR
- Ensure **no rotation** of thorax by placing the midcoronal plane parallel to the IR
- Raise or lower CR and IR as needed to the level of T7 for an average patient (top of IR is approximately 1½ to 2 inches [4 to 5 cm] above shoulders on average patients)

CR

- CR perpendicular to IR and centered to **midsagittal plane at level of T7** (7 to 8 inches [18 to 20 cm] below vertebra prominens, or to the inferior angle of scapula)
- IR centered to CR

Recommended Collimation Collimate on four sides to area of lung fields (top border of illuminated field should be to the level of vertebra prominens, and lateral border should be to outer skin margins)

Respiration Exposure is made at end of **second full inspiration**.

NOTE 1: For hypersthenic and broad-chested patients, place 14 × 17 inch (35 × 43 cm) IR landscape (or collimate to at least this size when possible with larger digital plates).

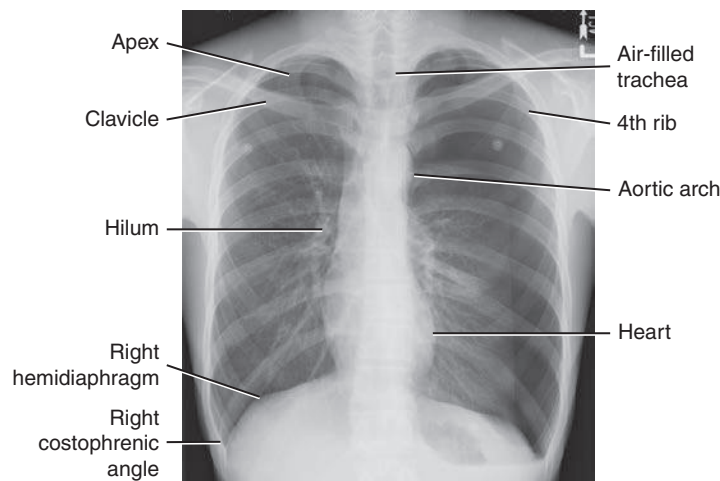


Fig. 2.54 PA chest.

Evaluation Criteria

Anatomy Demonstrated: • Included are both lungs from apices to costophrenic angles and the air-filled trachea from T1 down. • Hilum region markings, heart, great vessels, and bony thorax are demonstrated (Figs. 2.53 and 2.54).

Position: • Chin sufficiently elevated to prevent superimposing apices. • Sufficient forward shoulder rotation to prevent superimposition of scapulae over lung fields. • Larger breast shadows (if present) primarily lateral to lung fields. • **No rotation:** Both sternoclavicular joints the same distance from center line of spine.⁴ • Distance from lateral rib margins to vertebral column the same on each side from upper to lower rib cage (see NOTE 2). • Collimation margins near equal on top and bottom with center

of collimation field (CR) to T7 region on most patients. • Full inspiration with no motion. • Visualizes a minimum of 10 posterior ribs above diaphragm (11 on many patients).

NOTE 2: Scoliosis and kyphosis also may cause asymmetry of sternoclavicular joints and rib cage margins, as evidenced by R to L spinal curvature.

Exposure: • No motion, as evidenced by sharp outlines of rib margins, diaphragm, and heart borders and also sharp lung markings in hilar region and throughout lungs. • Sufficient long-scale contrast for visualization of fine vascular markings within lungs. • Faint outlines of at least midthoracic and upper thoracic vertebrae and posterior ribs visible through heart and mediastinal structures.

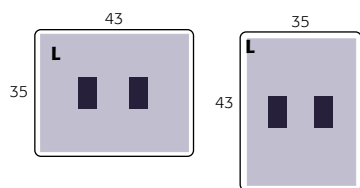
PA PROJECTION—CHEST**ON STRETCHER IF PATIENT CANNOT STAND****Clinical Indications**

- When performed erect, PA demonstrates pleural effusion, pneumothorax, atelectasis, and signs of infection

Chest
ROUTINE
• PA
• Lateral

Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait or landscape
- Grid
- kVp range—110–125



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

- Patient erect, seated on cart, legs over the edge (Fig. 2.55)
- Arms around cassette unless a chest IR device is used, then position as for an ambulatory patient
- Shoulders rotated forward and downward

Part Position

- Ensure no rotation of thorax.
- Adjust height of IR so that top of IR is about 1 ½ to 2 inches (4 to 5 cm) above top of shoulders and CR is at T7.
- If portable image receptor is used because patient cannot be placed up against wall bucky, place pillow or padding on lap to raise and support image receptor as shown, but keep it against chest for minimum object–image receptor distance (OID) (Fig. 2.56).

CR

- CR perpendicular to the IR and centered to the **midsagittal plane at the level of T7** (7 to 8 inches [18 to 20 cm] below vertebra prominens or to the inferior angles of scapulae)
- Cassette centered to level of CR

Recommended Collimation Collimate to area of lung fields. Upper border of illuminated field should be to the **level of vertebra prominens**, which with divergent rays will result in upper collimation border on IR to about 1 ½ inches (3.5 cm) above apex of lungs

Respiration Make exposure on second full inspiration.

NOTE: Use a compression band or other means to ensure that patient is stable and will not waver or move during exposure.

Evaluation Criteria

Radiograph should appear similar to ambulatory PA chest, as described on preceding page. (Fig. 2.57).

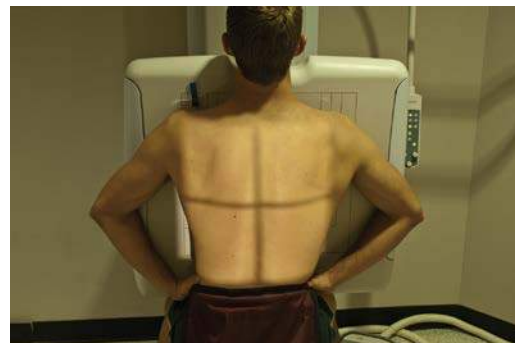


Fig. 2.55 PA chest (patient seated, chest against wall bucky).

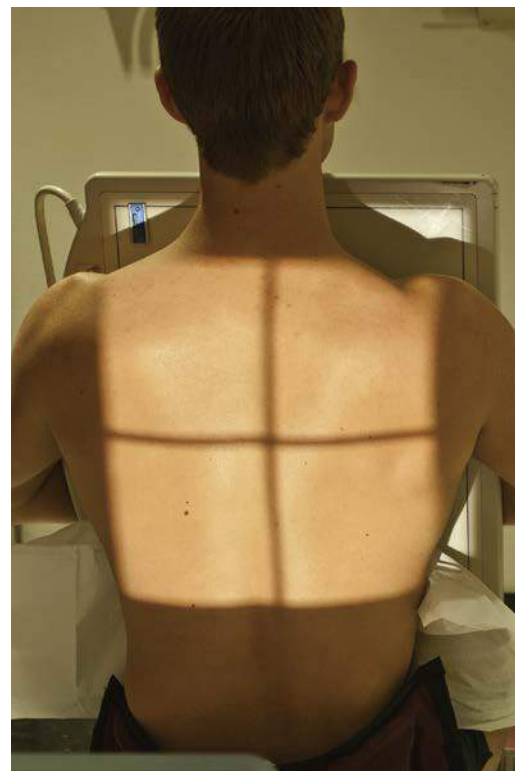


Fig. 2.56 PA chest (patient seated, holding cassette-less detector).



Fig. 2.57 PA chest.

LATERAL POSITION—CHEST

AMBULATORY PATIENT

Clinical Indications

- A 90° perspective from PA projection may demonstrate pathology situated posterior to the heart, great vessels, and sternum

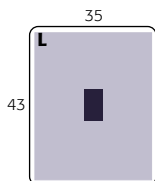
Chest

ROUTINE

- PA
- Lateral

Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid
- kVp range—110–125



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

- Patient erect, **left side** against IR unless patient complaint involves right side (in that case, do a right lateral if departmental protocol includes this option)
- Weight evenly distributed on both feet
- Arms raised above head, chin up

Part Position

- Center patient to CR and to IR anteriorly and posteriorly (Fig. 2.58)
- Position in a **true lateral** position (coronal plane is perpendicular and sagittal plane is parallel to IR; see NOTE 1)
- Lower CR and IR slightly from PA if needed (see NOTE 2)

CR

- CR perpendicular, directed to **midthorax at level of T7** (3 to 4 inches [7.5 to 10 cm] below level of jugular notch)

Recommended Collimation Collimate on four sides to area of lung fields (top border of light field to level of vertebra prominens).

Respiration Make exposure at end of **second full inspiration**.

NOTE 1: Ensure that **midsagittal plane** is **parallel to IR**, which for slender but broad-shouldered patients results in hips and lower thorax *not* being against IR.

NOTE 2: This increase in OID of the lower chest results in the costophrenic angles of the lungs being projected lower because of divergence of the x-ray beam. Therefore, **CR and IR should be lowered a minimum of 1 inch (2.5 cm) from the PA** on this type of patient to prevent cutoff of costophrenic angles.

Evaluation Criteria

Anatomy Demonstrated: • Included are the entire lungs from apices to the costophrenic angles and from the sternum anteriorly to the posterior ribs and thorax posteriorly (Figs. 2.59 and 2.60).

Position: • Chin and arms elevated sufficiently to prevent excessive soft tissues from superimposing apices. • **No rotation:** Posterior ribs and costophrenic angle on side away from IR projected slightly ($\frac{1}{4}$ to $\frac{1}{2}$ inch [or about 1 cm] posterior because of divergent rays). • The hilar region should be in the approximate center of the IR.

NOTE: To determine direction of rotation and critique radiographs, see p. 14.

Exposure: • No motion, as evidenced by sharp outlines of the diaphragm and lung markings. • Should have sufficient exposure and long-scale contrast for **visualization of rib outlines and lung markings through the heart shadow and upper lung areas** without overexposing other regions of the lungs.

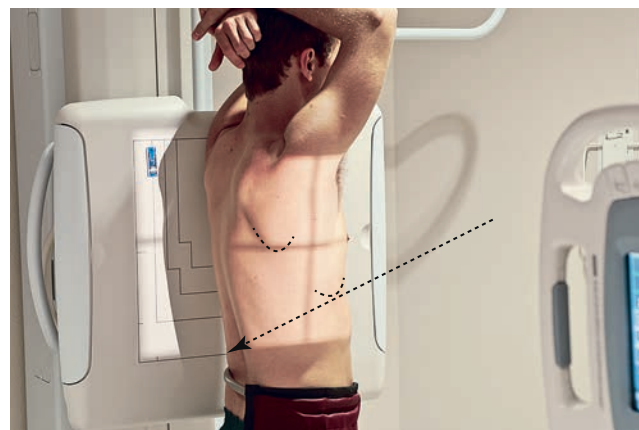


Fig. 2.58 Left lateral chest position.



Fig. 2.59 Left lateral chest.

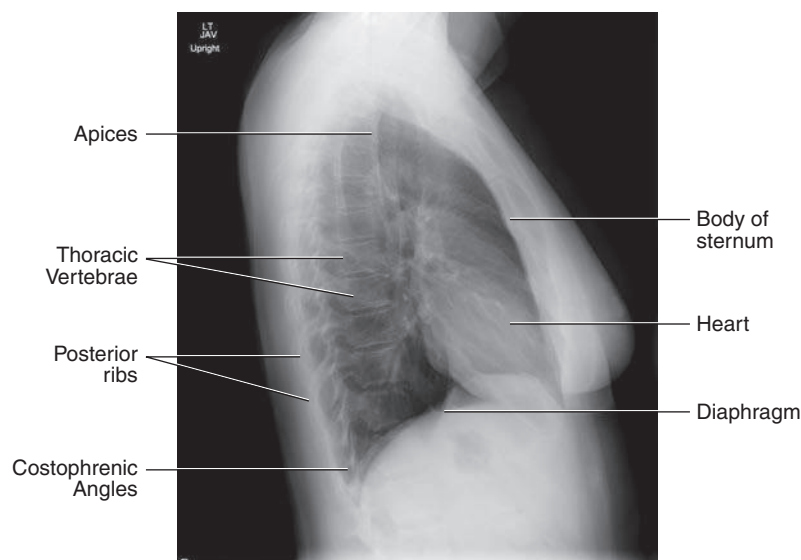


Fig. 2.60 Lateral chest.

ALTERNATIVE LATERAL POSITIONS—CHEST

WITH WHEELCHAIR OR CART IF PATIENT CANNOT STAND

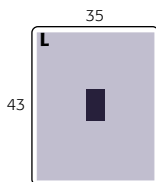
Clinical Indications

- A 90° perspective from PA may demonstrate pathology situated posterior to the heart, great vessels, and sternum

Chest
ROUTINE
• PA
• Lateral

Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid
- kVp range—110–125



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position on Cart

- Patient seated on cart; legs over the edge if this is easier for patient (ensure that cart is locked and does not move)
- Arms crossed above head or hold on to arm support (Fig. 2.61)
- Chin extended upward

Patient Position in Wheelchair

- Remove armrests, if possible, or place pillow or other support under smaller patients so that armrests of wheelchair do not superimpose lower lungs. (Fig. 2.62).
- Turn patient in wheelchair to lateral position as close to IR as possible.
- Have patient lean forward and place support blocks behind back; raise arms above head and have patient hold on to support bar—keeping arms high.

Part Position

- Center patient to CR and to IR by checking anterior and posterior aspects of thorax; adjust CR and IR to level of T7.
- Ensure **no rotation** by viewing patient from tube position.

CR

- CR perpendicular, directed to **level of T7** (3 to 4 inches [8 to 10 cm] below level of jugular notch)
- Top of IR approximately 1 inch (2.5 cm) above vertebra prominens

Recommended Collimation Collimate on four sides to area of lung fields (top border of light field to level of vertebra prominens).

Respiration Make exposure at end of **second full inspiration**.

NOTE: Always attempt to have patient sit completely erect in wheelchair or on cart if possible. However, if the patient's condition does not allow this, the head end of the cart can be raised as nearly erect as possible with a radiolucent support behind the back (Fig. 2.63). All attempts should be made to get patient as nearly erect as possible.



Fig. 2.61 Left lateral chest position on cart.



Fig. 2.62 Left lateral position in wheelchair (arms up, support behind back).

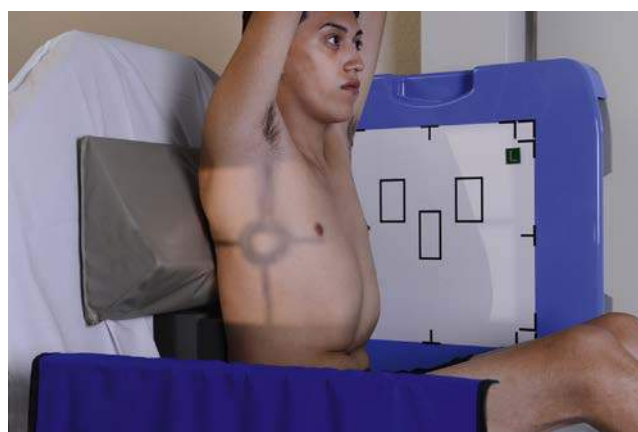


Fig. 2.63 Erect, supported left lateral position.

Evaluation Criteria

Radiograph should appear similar to ambulatory lateral position as described under Evaluation Criteria on preceding page.

AP PROJECTION—CHEST

SUPINE OR SEMIERECT (IN DEPARTMENT OR AS BEDSIDE PORTABLE)

Clinical Indications

- Demonstrates pathology involving the lungs, diaphragm, and mediastinum
- Determining air-fluid levels (pleural effusion) requires a completely erect position with a horizontal CR, as in a PA or decubitus chest projection

Chest**SPECIAL**

- AP supine or semierect

Technical Factors

- Minimum SID—72 inches (180 cm) for semierect (see NOTES)
- IR size—14 × 17 inches (35 × 43 cm), portrait or landscape
- Grid (due to higher kVp, the use of a grid is strongly recommended)
- kVp range—110–125

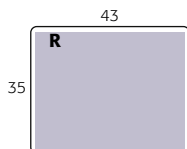


Fig. 2.64 AP supine.

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

- Patient is supine on cart; if possible, the head end of the cart or bed should be raised into a semierect position (see NOTES).
- Roll patient's shoulders forward by rotating arms medially or internally.

Part Position

- Place IR under or behind patient; align center of IR to CR (top of IR about 1½ inches [4 to 5 cm] above shoulders) (Fig. 2.64).
- Center patient to CR and to IR; check by viewing patient from the top, near the tube position.

CR

- CR angled **caudad to be perpendicular to long axis of sternum** (generally requires $\pm 5^\circ$ caudad angle, to prevent clavicles from obscuring the apices)
- CR to **level of T7**, 3 to 4 inches (8 to 10 cm) below jugular notch

Recommended Collimation Collimate on four sides to area of lung fields (top border of light field to level of vertebra prominens).

Respiration Make exposure at end of **second full inspiration**.

NOTES: Crosswise IR placement is recommended for large or hypersthenic or broad-chested patients to minimize chance of lateral cutoff. This requires **accurate CR alignment with center of IR** with only minimal caudad angle to prevent grid cutoff if grid is used.

For **semierect position**, use 72-inch (180-cm) SID if possible. **Always** place markers on the IR or label the image to indicate the SID used; also indicate the projections obtained, such as AP supine or AP semierect (Fig. 2.65).

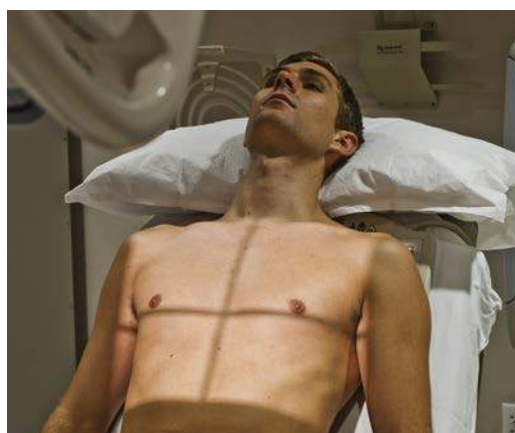


Fig. 2.65 AP semierect.



Fig. 2.66 AP semierect.

Evaluation Criteria

- Criteria for chest radiographs taken in supine or semierect positions should be similar to criteria for PA projection described earlier, with three exceptions:
 1. The heart appears larger as a result of increased magnification from a shorter SID and increased OID of the heart.
 2. Possible pleural effusion for this type of patient often obscures vascular lung markings compared with a fully

erect PA chest projection. Without a horizontal beam, fluid levels may not be demonstrated.

3. Usually, inspiration is not as full, and only eight or nine posterior ribs are visualized above the diaphragm. The lungs may appear denser because they are not as fully aerated (Fig. 2.66).
- **Correct CR angle:** The clavicles should be in the same horizontal plane with an unobstructed view of the apical region.⁴

LATERAL DECUBITUS POSITION (AP PROJECTION)—CHEST

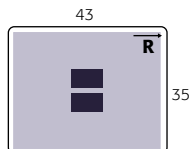
Clinical Indications

- Small **pleural effusions** are demonstrated by air-fluid levels in pleural space
- **Small amounts of air** in pleural cavity may demonstrate a possible pneumothorax (see NOTES)

Chest
SPECIAL
• AP supine or semierect
• Lateral decubitus (AP)

Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape (with respect to patient position)
- Grid
- kVp range—110–125
- Use decubitus (decub) marker



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

- Cardiac board on the cart or radiolucent pad under patient
- Patient lying on right side for right lateral decubitus and on left side for left lateral decubitus (see NOTES)
- Patient's chin extended and both arms raised above head to clear lung field; back of patient firmly against IR; cart secured to prevent patient from moving forward and possibly falling; pillow under patient's head (Fig. 2.67)
- Knees flexed slightly and coronal plane parallel to IR with **no body rotation**

Part Position

- Adjust height of IR to center thorax to IR (see NOTES)
- Adjust patient and cart to center midsagittal plane and T7 to CR (top of IR is approximately 1 inch [2.5 cm] above vertebra prominens)

CR

- CR horizontal, directed to center of IR, to **level of T7**, 3 to 4 inches (8 to 10 cm) inferior to level of jugular notch. A **horizontal beam must be used** to show air-fluid level or pneumothorax.

Recommended Collimation Collimate on four sides to area of lung fields (top border of light field to level of vertebra prominens) (see NOTES).

Respiration Make exposure at end of **second full inspiration**.

Alternative Positioning Some department protocols state that the head should be 10° lower than the hips to reduce the apical lift caused by the shoulder, allowing the entire chest to remain horizontal (requires support under hips).

NOTES: Place appropriate decubitus marker and R or L to indicate which side of chest is down.

Radiograph may be taken as a right or left lateral decubitus. To produce the most diagnostic images, both lungs should be included on the image. For **possible fluid** in the pleural cavity (pleural effusion), the suspected side should be **down**. Do *not* cut off that side of the chest. The anatomic side marker must correspond with the patient's left or right side of the body. The marker must be placed on the IR before exposure. It is unacceptable practice to indicate the side of the body either digitally or with a marking pen after the exposure.

For possible **small amounts of air** in the pleural cavity (pneumothorax), the affected side should be **up**, and care must be taken *not* to cut off this side of the chest.

Evaluation Criteria

Anatomy Demonstrated: • Entire lungs, including apices, both costophrenic angles, and both lateral borders of ribs, should be included (Figs. 2.68 and 2.69).

Position: • **No rotation:** Should show equal distance from the vertebral column to the lateral borders of the ribs on both sides; sternoclavicular joints should be the same distance from the vertebral column. • Arms should not superimpose upper lungs. • Collimation field (CR) should be centered to the area of T7 on average-sized patients.

Exposure: • **No motion;** diaphragm, rib, and heart borders and lung markings should appear sharp. • Optimal contrast scale and exposure should result in **faint visualization of vertebrae and ribs through heart shadow**.



Fig. 2.67 Left lateral decubitus position (AP projection).

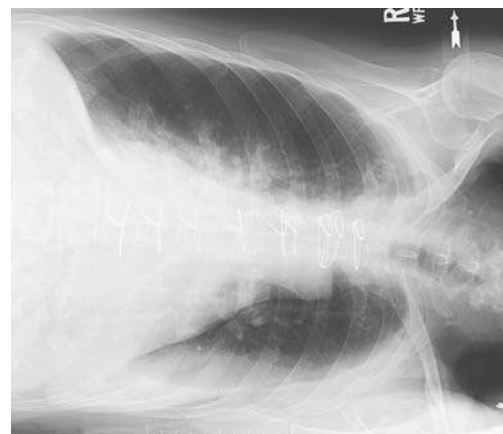


Fig. 2.68 Left lateral decubitus (fluid evident in left lung).

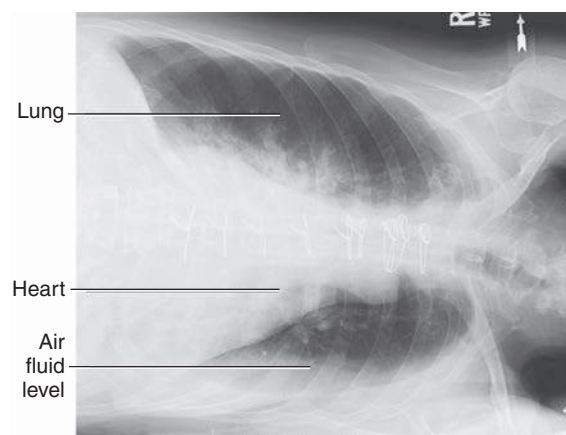


Fig. 2.69 Left lateral decubitus.

AP LORDOTIC PROJECTION—CHEST

Clinical Indications

- Rule out calcifications and masses beneath the clavicles

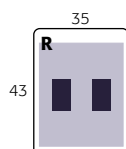
Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait or landscape
- Grid
- kVp range—110–125

Chest

SPECIAL

- AP supine or semierect
- Lateral decubitus (AP)
- AP lordotic



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

- Patient standing about 1 foot (30 cm) away from IR and leaning back with shoulders, neck, and back of head against IR
- Both patient's hands on hips, palms out; shoulders rolled forward (Fig. 2.70)

Part Position

- Center midsagittal plane to CR and to centerline of IR.
- Center cassette to CR (top of IR should be about 3 inches [7 to 8 cm] above shoulders on an average patient).
- Palpate clavicles to ensure they are at level or above shoulders.

CR

- CR **perpendicular** to IR, centered to **midsternum** (3 to 4 inches [9 cm] below jugular notch)

Recommended Collimation Collimate on four sides to area of lung fields (top border of light field to level of vertebra prominens).

Respiration Make exposure at end of **second full inspiration**.

Alternative Lordotic Projection If patient is weak and unstable or is unable to assume the erect lordotic position, an AP semiaxial projection may be taken with the patient in a supine position (Fig. 2.71). Shoulders are rolled forward and arms positioned as for lordotic position. The CR is directed **15° to 20° cephalad**, to the midsternum.

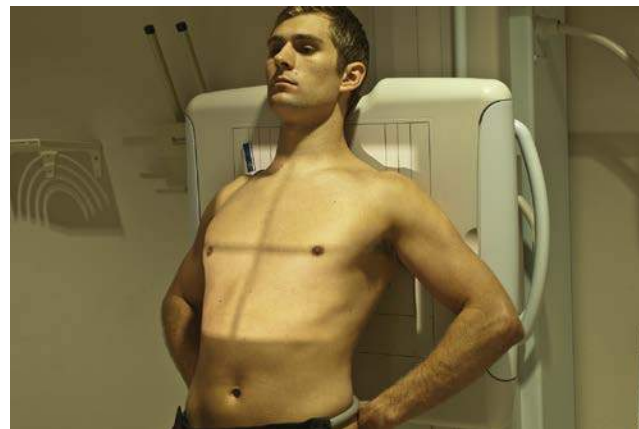


Fig. 2.70 AP lordotic position.



Fig. 2.71 Alternative: semiaxial AP.



Fig. 2.72 AP lordotic.

Evaluation Criteria

Anatomy Demonstrated: • Entire lung fields and clavicles should be included (Fig. 2.72).

Position: • Clavicles should appear nearly horizontal and **above or superior** to apices, with medial aspects of clavicles superimposed by first ribs. • Ribs appear distorted, with posterior ribs appearing nearly horizontal and superimposing anterior ribs. • **No rotation:** Sternal ends of the clavicles should be the same distance from the vertebral column on each side. The lateral borders of the ribs on both sides should appear to be at nearly equal distances from the vertebral column. • Center of collimation field (CR) should be midsternum with collimation visible on top and bottom.

Exposure: • No motion; diaphragm, heart, and rib outlines should appear sharp. • Optimal contrast scale and exposure should allow visualization of the faint vascular markings of lungs, especially in area of the apices and upper lungs.

ANTERIOR OBLIQUE POSITIONS: RAO AND LAO—CHEST

Clinical Indications

- Investigate pathology involving the lung fields, trachea, and mediastinal structures
- Determine the size and contours of the heart and great vessels

Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid
- kVp range—110–125

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

- Patient erect, rotated 45° with right anterior shoulder against IR for RAO (Fig. 2.73) and 45° with left anterior shoulder against IR for LAO (Fig. 2.74) (see NOTES for 60° LAO)
- Patient's arm flexed nearest IR and hand placed on hip, palm out
- Opposite arm raised to clear lung field and hand rested on head or on chest unit for support, keeping arm raised as high as possible
- Patient looking straight ahead; chin raised

Part Position

As viewed from the x-ray tube, center the patient to CR and to IR, with top of IR about 1 inch (2.5 cm) above vertebra prominens.

CR

- CR perpendicular, directed to **level of T7** (7 to 8 inches [8 to 10 cm] below level of vertebra prominens)
- CR midway between midsagittal plane and lateral margin of thorax

Recommended Collimation Collimate on four sides to area of lung fields (top border of light field to level of vertebra prominens).

Respiration Make exposure at end of **second full inspiration**.

NOTES: For **anterior** oblique, the side of interest generally is the side **farthest** from the IR. Thus, the **RAO** provides the best visualization of the **left** lung.

Certain positions for studies of the heart and great vessels require oblique positions with an increase in rotation of 45° to 60° (see Figs. 2.75 and 2.76).

Less rotation (15° to 20°) may be valuable for better visualization of the various areas of the lungs for possible pulmonary disease (Fig. 2.77).

Exception Either erect or recumbent posterior oblique projections can be taken if the patient cannot assume an erect position for anterior oblique, or if supplementary projections are required.

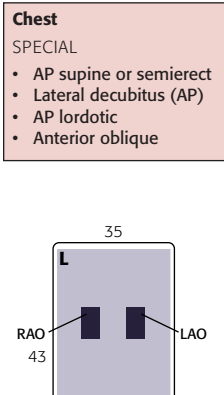


Fig. 2.73 45° RAO position.

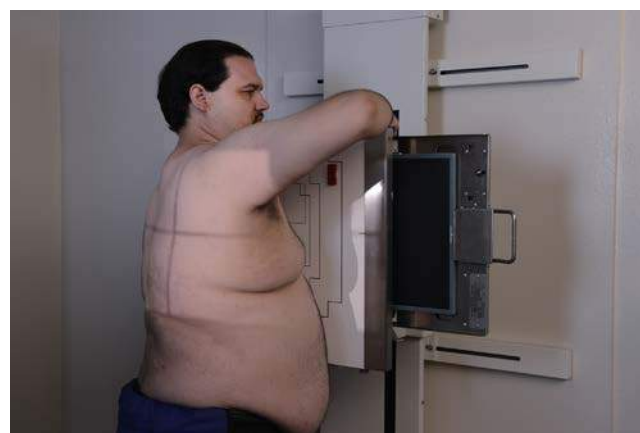


Fig. 2.74 45° LAO position.

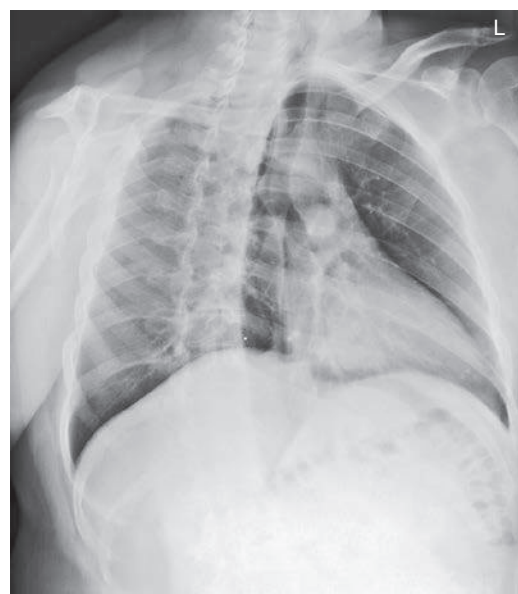


Fig. 2.75 45° RAO position.

Evaluation Criteria

Anatomy Demonstrated: • Both lungs from the apices to the costophrenic angles should be included. • Air-filled trachea, great vessels, and heart outlines are best visualized with 60° LAO position.

Position: • To evaluate for a 45° rotation, the distance from the outer margin of the ribs to the vertebral column on the side

farthest from the IR should be approximately two times the distance of the side closest to the IR (Figs. 2.78 and 2.79). • CR centered at level of T7.

Exposure: • No motion; outline of the diaphragm and heart should appear sharp. • Optimal exposure and contrast allow visualization of vascular markings throughout the lungs and rib outlines except through the densest regions of the heart.

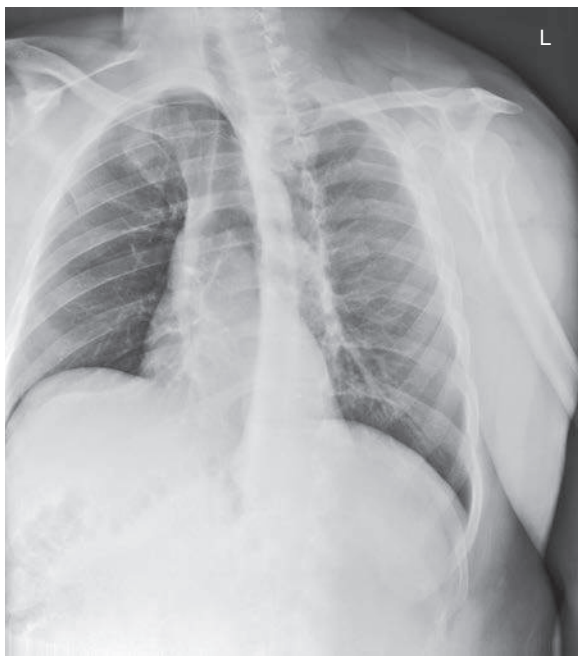


Fig. 2.76 45° LAO position.

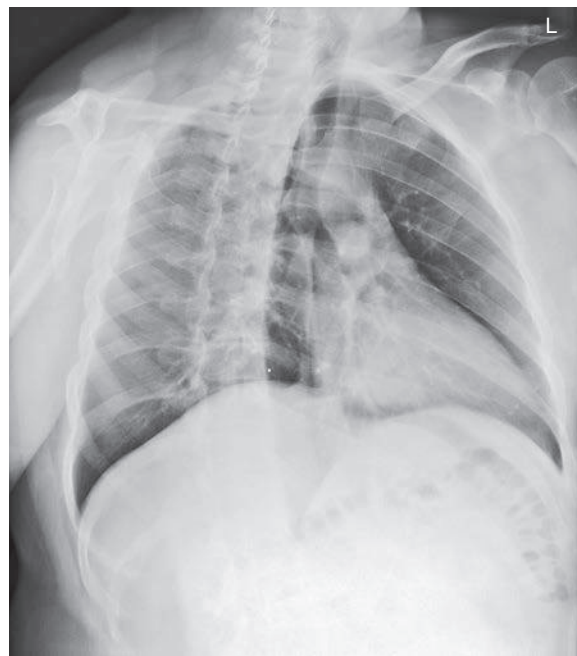


Fig. 2.78 45° RAO position.

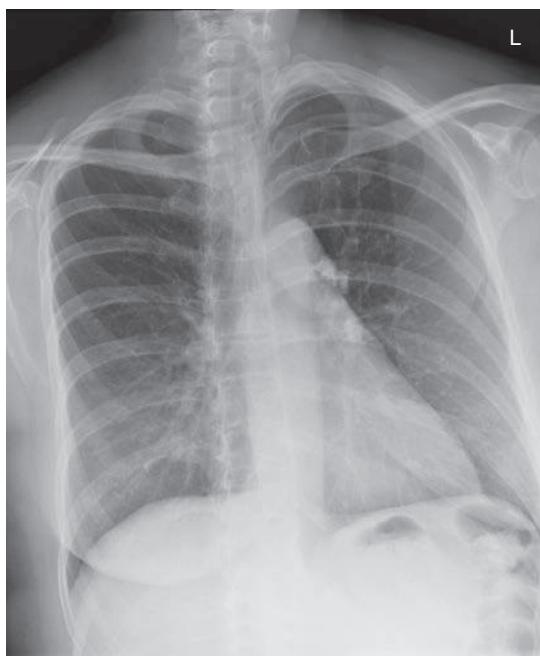


Fig. 2.77 15° to 20° RAO position.

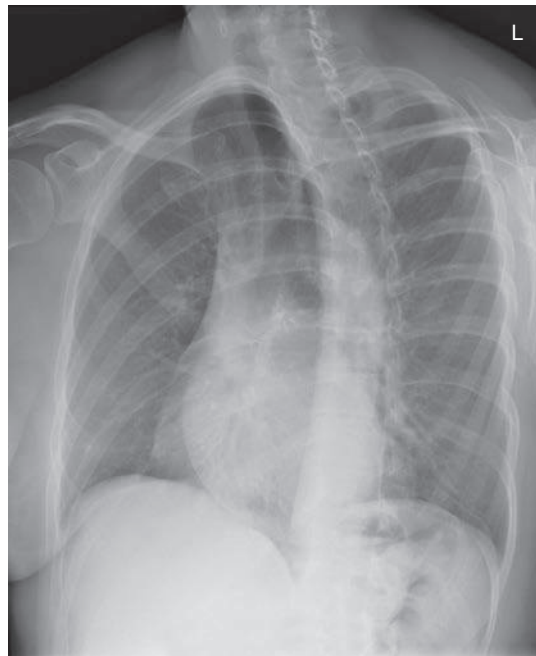


Fig. 2.79 45° LAO position.

POSTERIOR OBLIQUE POSITIONS: RPO AND LPO—CHEST

Clinical Indications

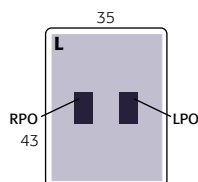
- Investigate pathology involving the lung fields, trachea, and mediastinal structures
- Determine the size and contours of the heart and great vessels

Technical Factors

- Minimum SID—72 inches (180 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid
- kVp range—110–125

Chest
SPECIAL

- AP supine or semierect
- Lateral decubitus (AP)
- AP lordotic
- Anterior oblique
- Posterior oblique



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position (Erect)

- Patient erect, rotated 45° (up to 60°) with right posterior shoulder against IR for RPO (Fig. 2.80) and 45° (up to 60°) with left posterior shoulder against IR for LPO (Fig. 2.81)
- Arm closest to the IR raised, resting on head; other arm placed on hip with palm out
- Patient looking straight ahead

Patient Position (Recumbent)

- If patient cannot stand or sit, perform posterior oblique projections on table.
- Place supports under patient's head and under elevated hip and shoulder.

Part Position

- Top of IR about 1 inch (2 cm) above vertebra prominens or about 5 inches (12 cm) above level of jugular notch (2 inches [5 cm] above shoulders)
- Thorax centered to CR and to IR

CR

- CR perpendicular, to level of T7
- CR midway between midsagittal plane and lateral margin of thorax

Recommended Collimation Collimate on four sides to area of lung fields (top border of light field to level of vertebra prominens).

Respiration Make exposure after second full inspiration.

NOTES: Posterior oblique projections provide best visualization of the side closest to the IR.

Posterior positions show the same anatomy as the opposite anterior oblique positions. Thus, the RPO (Fig. 2.82) corresponds to the LAO position and the LPO (Fig. 2.83) corresponds to the RAO position.

Evaluation Criteria

Evaluation criteria are similar to criteria for anterior oblique positions described earlier. • However, because of increased magnification of the anterior diaphragm, lung fields usually appear shorter on posterior oblique than on anterior oblique projections. • The heart and great vessels also appear larger on posterior oblique because they are farther from the IR (Figs. 2.84 and 2.85).



Fig. 2.80 45° RPO position.



Fig. 2.81 45° LPO position.

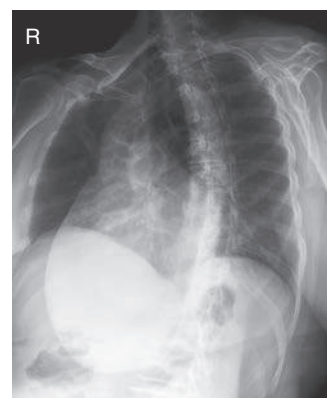


Fig. 2.82 45° to 60° RPO position. **Fig. 2.83** 45° to 60° LPO position.

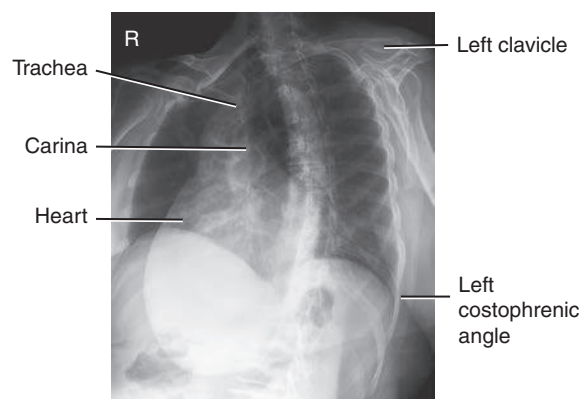


Fig. 2.84 45° to 60° RPO position.

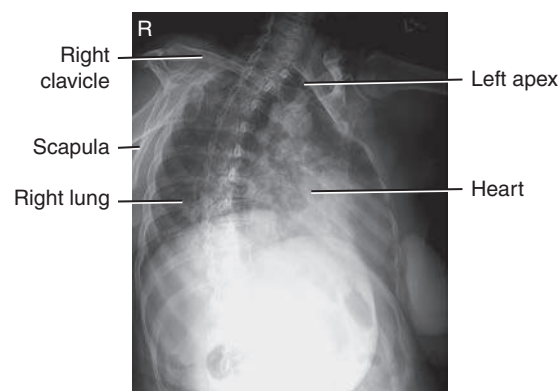


Fig. 2.85 45° to 60° LPO position.

LATERAL POSITION—UPPER AIRWAY

Clinical Indications

- Investigate pathology of the air-filled larynx and trachea, including the region of thyroid and thymus glands and upper esophagus, for opaque foreign object or if contrast medium is present
- Rule out **epiglottitis**, which may be life-threatening for a young child

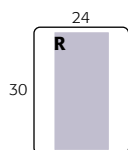
Upper Airway

ROUTINE

- Lateral
- AP

Technical Factors

- Minimum SID—72 inches (180 cm) to minimize magnification
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—75–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Patient should be upright if possible, seated, or standing in a lateral position (may be taken in R or L lateral and may be taken recumbent tabletop if necessary).

Part Position

- Position patient to center upper airway to CR and to center of IR (larynx and trachea lie anterior to cervical and thoracic vertebrae).
- Rotate shoulders posteriorly with arms hanging down and hands clasped behind back.
- Raise chin slightly and have patient look directly ahead (Fig. 2.86).
- Adjust IR height to place top of IR at level of external auditory meatus (EAM), which is the opening of the external ear canal (see Respiration if area of primary interest is the trachea rather than the larynx).

CR

- CR perpendicular to center of IR at **level of C6 or C7**, midway between the laryngeal prominence of the thyroid cartilage and the jugular notch

Recommended Collimation Collimate to region of soft tissue of the neck.

Respiration Make exposure **during a slow, deep inspiration** to ensure filling trachea and upper airway with air.

NOTE (centering and exposure for neck region): Centering should be to laryngeal prominence (C5) with exposure factors for a soft tissue lateral neck if the area of interest is primarily the larynx and upper trachea.

NOTE (centering and exposure for distal larynx and trachea region): If the distal larynx and upper trachea and midtrachea are the primary areas of interest, the IR and CR should be lowered to place the CR at the upper jugular notch (T1–T2) with exposure factors approximately those for a lateral chest.

Evaluation Criteria

Anatomy Demonstrated: • The larynx and trachea should be filled with air and well visualized (Figs. 2.87 and 2.88).

Position: • Centering for the neck region (larynx and proximal trachea) should include the EAM at the upper border of the image and T2 or T3 on the lower border. If the **distal larynx and trachea** is the primary area of interest, centering should be lower to include the area from C3 to T4 or T5 on the image. • The shadows of the shoulders should be primarily posterior to and should not superimpose the area of the trachea. • Collimation borders should appear on both sides with ideally only minimal ($\leq 1/4$ inch) borders on top and bottom.

Exposure: • Optimal exposure includes a soft tissue technique wherein the air-filled larynx and upper trachea are not overexposed. • Cervical vertebrae appear underexposed.

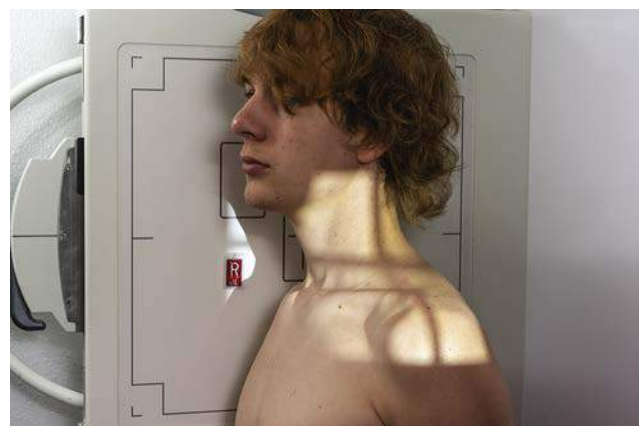


Fig. 2.86 Right lateral position—upper airway.



Fig. 2.87 Lateral—upper airway (for distal larynx and trachea region).

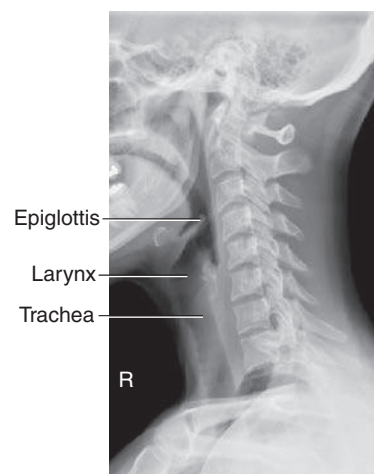
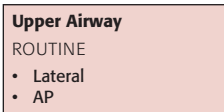


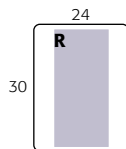
Fig. 2.88 Lateral—upper airway.

AP PROJECTION—UPPER AIRWAY**Clinical Indications**

- Investigate pathology of the air-filled larynx and trachea, including the region of the thyroid and thymus glands and upper esophagus for opaque foreign object or if contrast medium is present

**Technical Factors**

- Minimum SID—40 inches (100 cm) if possible, to minimize magnification
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—75–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Patient should be upright if possible, seated or standing with back of head and shoulders against IR (may be taken recumbent tabletop if necessary)

Part Position

- Align midsagittal plane with CR and with midline of grid or table.
- Raise chin so that **acanthiomeatal line is perpendicular to the IR** (line from the acanthion or area directly under the nose and the meatus or EAM); have patient look directly ahead (Fig. 2.89)

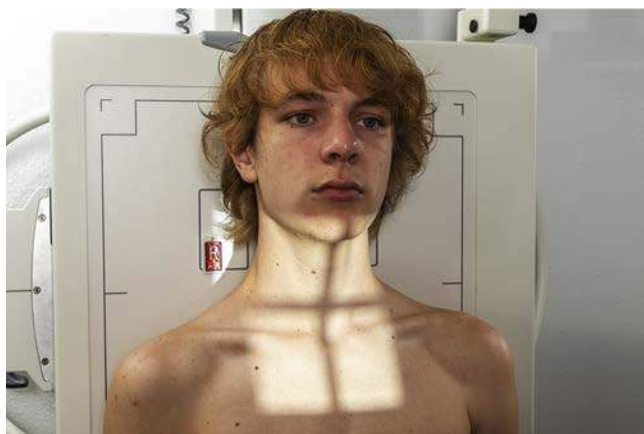


Fig. 2.89 AP—upper airway.

- Adjust the IR height to place top of IR about 1 or 1½ inches (3 to 4 cm) below EAM (see NOTE for explanation of centering)

CR

- CR perpendicular to center of IR at **level of T1–T2**, about 1 inch (2.5 cm) above the jugular notch

Recommended Collimation Collimate to region of soft tissue neck.

Respiration Make exposure **during a slow, deep inspiration** to ensure filling of trachea and upper airway with air.

NOTE (exposure): Exposure for this AP projection should be approximately that of an AP of the cervical or thoracic spine.

NOTE (centering for upper airway and trachea): Centering for this AP projection is similar to that of the lateral distal larynx and upper trachea position described on the previous page because the most proximal area of the larynx is not visualized on the AP as a result of the superimposed base of the skull and mandible. Therefore, more of the trachea can be visualized.

Evaluation Criteria

Anatomy Demonstrated: • The larynx and trachea from C3 to T4 should be filled with air and visualized through the spine. • The area of the proximal cervical vertebrae (the lower margin of the shadow of the superimposed mandible and base of skull) to the midthoracic region should be included (Fig. 2.90).

Position (see previous Notes): • **No rotation** should occur, as evidenced by the symmetric appearance of the sternoclavicular joints. • The mandible should superimpose the base of the skull with the spine aligned with the center of the film. • Collimation borders should appear on both sides with ideally only minimal ($\leq 1/4$ inch) borders on top and bottom. • The collimation field (CR) should be centered to the area of T1–T2.

Exposure: • Optimal exposure should be just dark enough to allow visualization of the air-filled trachea through the cervical and thoracic vertebrae.

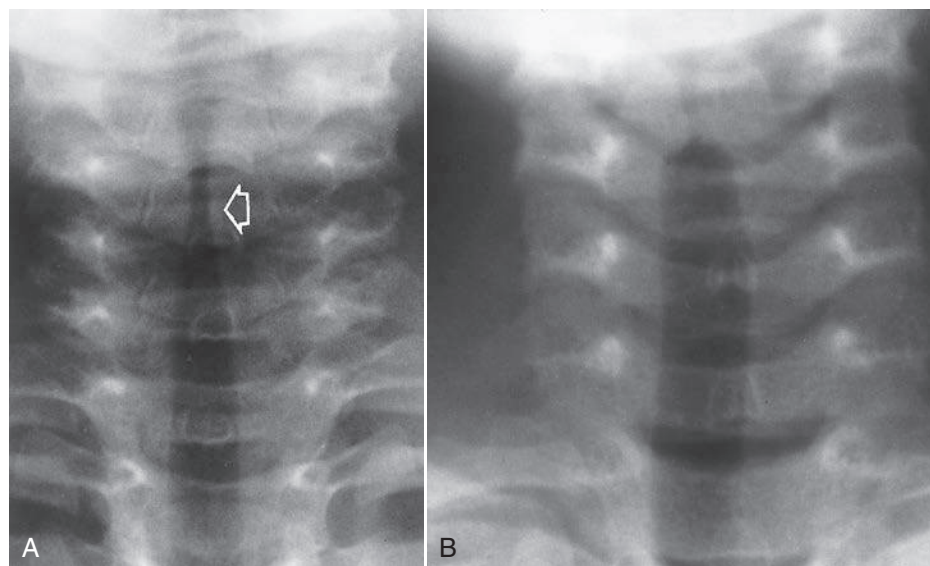


Fig. 2.90 Croup. A, Arrow indicates smooth, tapered narrowing of subglottic portion of trachea (Gothic arch sign). B, Normal trachea with broad shouldering in subglottic region. (From Eisenberg R, Johnson N: Comprehensive radiographic pathology, ed 5, St Louis, 2012, Mosby.)

RADIOGRAPHS FOR CRITIQUE

This section consists of an ideal projection (Image A) along with one or more projections that may demonstrate positioning and/or technical errors. Critique Figs C2.91 through C2.92. Compare Image A to the other projections and identify the errors. While examining each image, consider the following questions:

1. Is all essential anatomy demonstrated on the image?
2. What positioning errors are present that compromise image quality?

3. Are technical factors optimal?
 4. Is there evidence of collimation and pre-exposure anatomical side markers visible on the image?
 5. Do these errors require a repeat exposure?
- Feedback for each set of images is located on the faculty Evolve site.

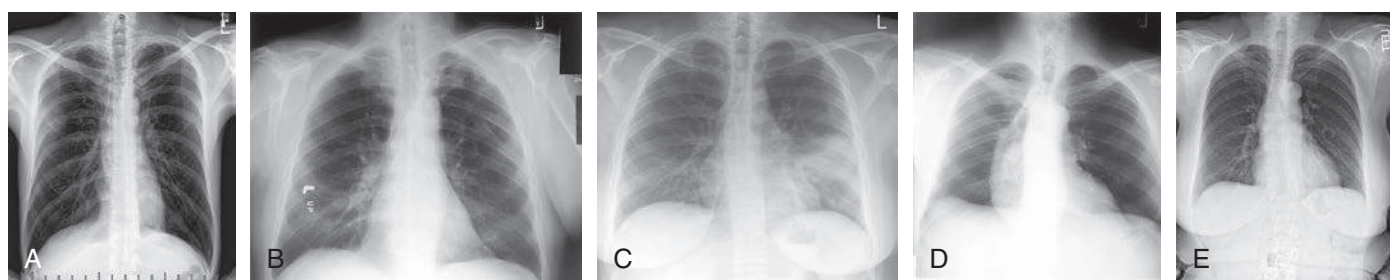


Fig. C2.91 PA chest.

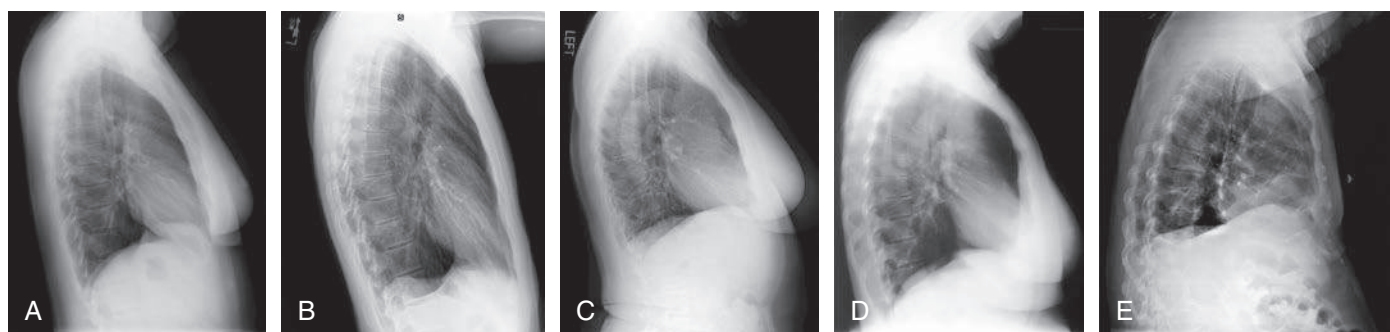


Fig. C2.92 Lateral chest.